

模拟电子技术基础

Analog Device

9 Signal Generation - Non sine-wave

信号的产生 – 非正弦信号

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电子与电气工程系 E³

9.5 Voltage comparator 电压比较器

Identify or compare the amplitude of input signals
用来鉴定或比较输入信号的幅值

Fundamental component of non-Sine-wave generator
非正弦发生器的基本电路

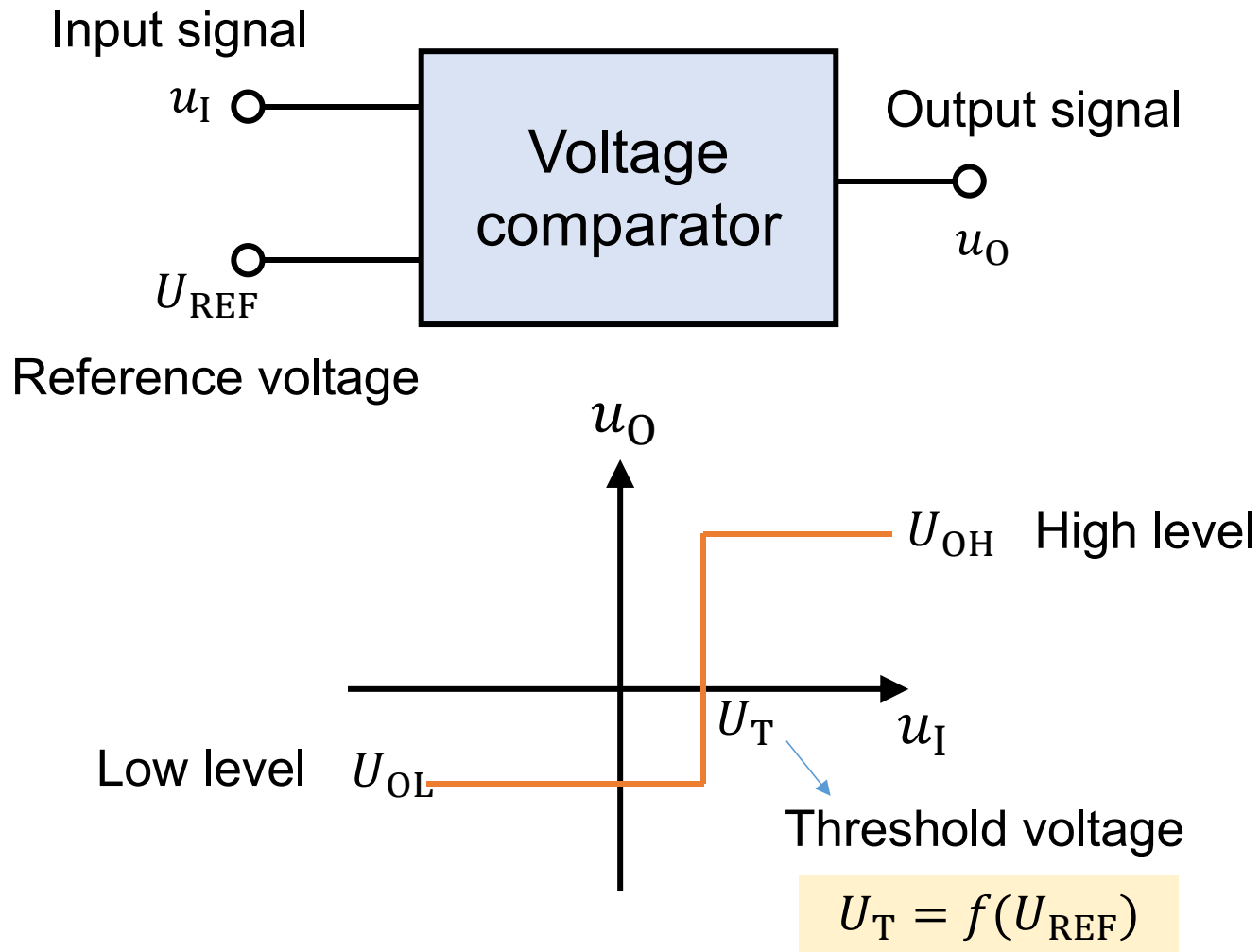
Voltage comparator 电压比较器

Input terminal {
Reference level/Voltage reference U_{REF} 参考电平/基准电压
Input voltage u_i 被比较的输入电压

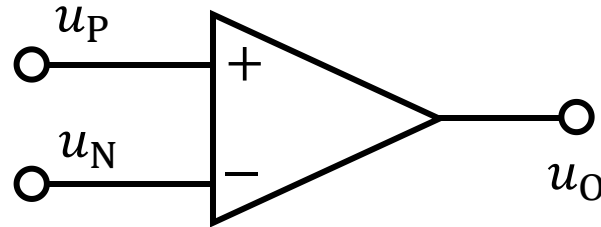
Output terminal: Output signal only has two values: high level U_{OH} or low level U_{OL} 输出信号只有高电平和低电平两个电压值

When input signal u_i crosses **threshold voltage U_T** ($U_T = f(U_{REF})$), output voltage will jump from U_{OH} to U_{OL} , or jump from U_{OL} to U_{OH} .
当输入信号达到阈值电压，输出电压将发生阶跃变化，从 U_{OH} 跃变为 U_{OL} 或从 U_{OL} 跃变为 U_{OH} 。

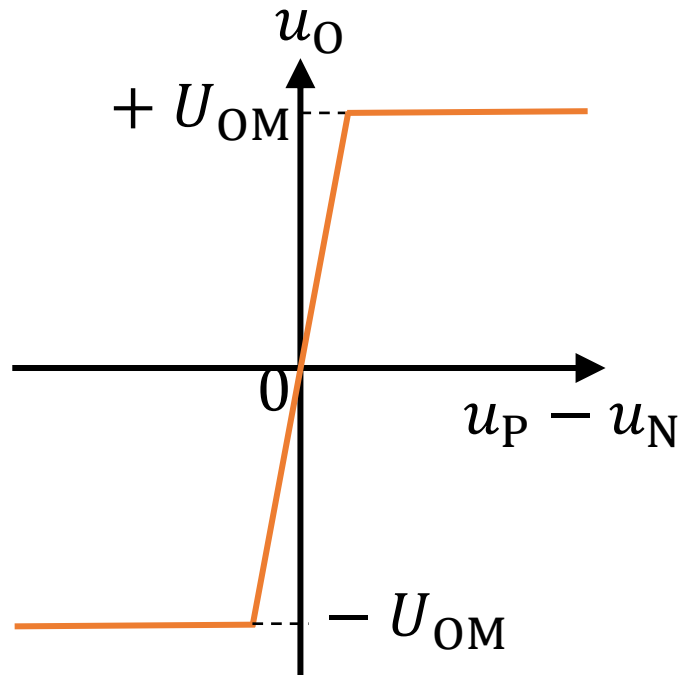
Voltage comparator 电压比较器



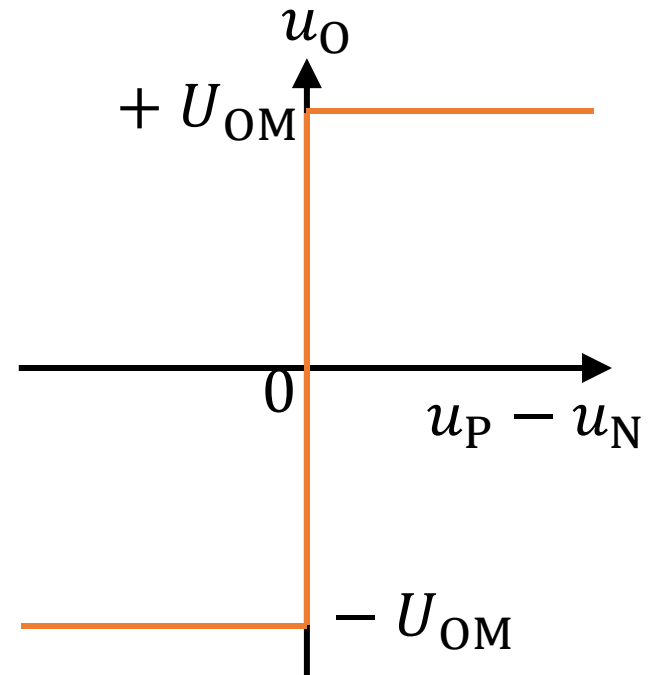
Real and ideal operational amplifier

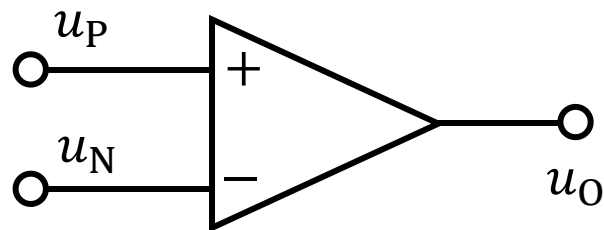


Real operational amplifier

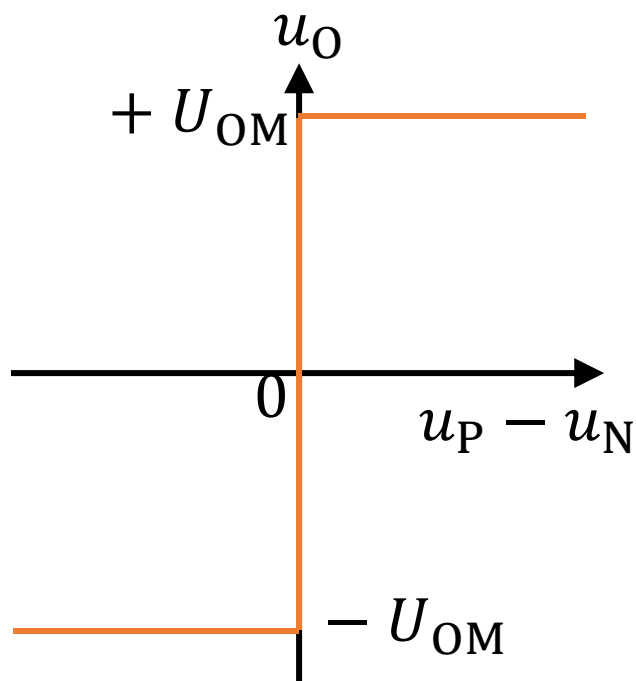


Ideal operational amplifier





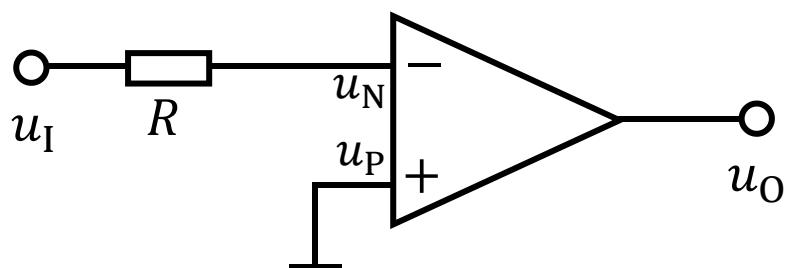
Ideal operational amplifier



In this chapter, we will use the saturation property of operational amplifier (运放的饱和特性) to realize voltage comparator and non-sine-wave generator.

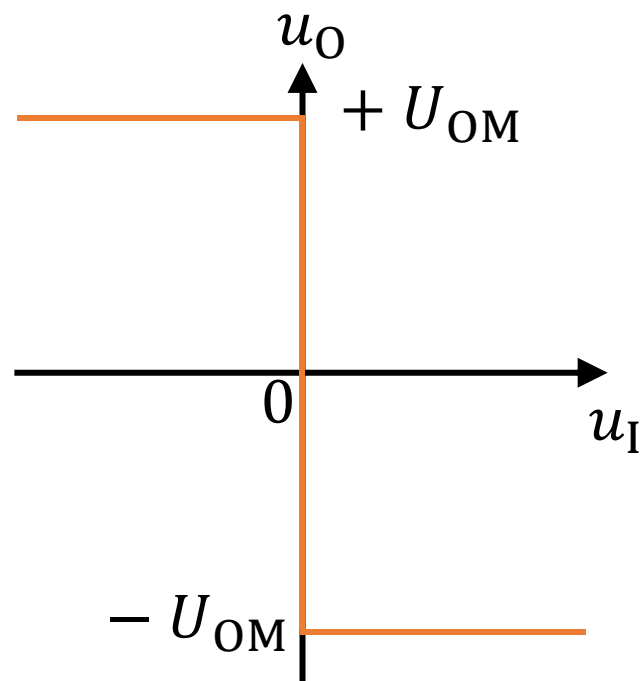
(1) Single-limit voltage comparator 单限比较器

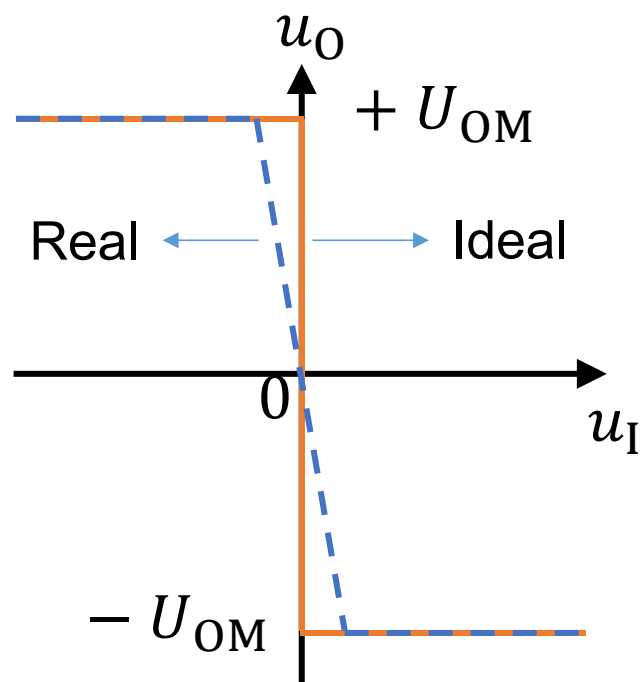
(a) Threshold voltage $U_T=0$

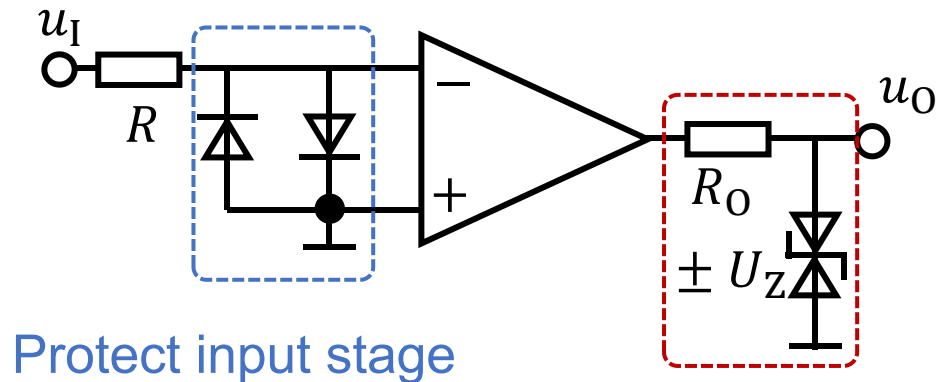
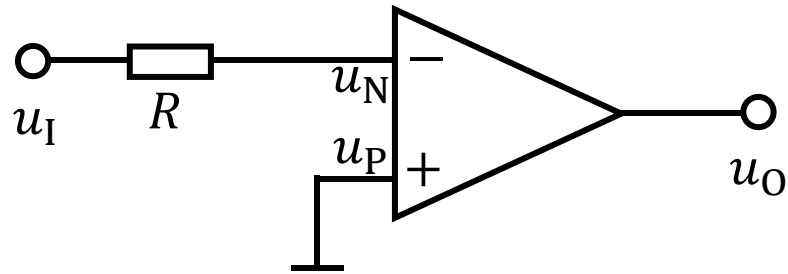


$$u_I = u_N = u_N - u_P = -(u_P - u_N)$$

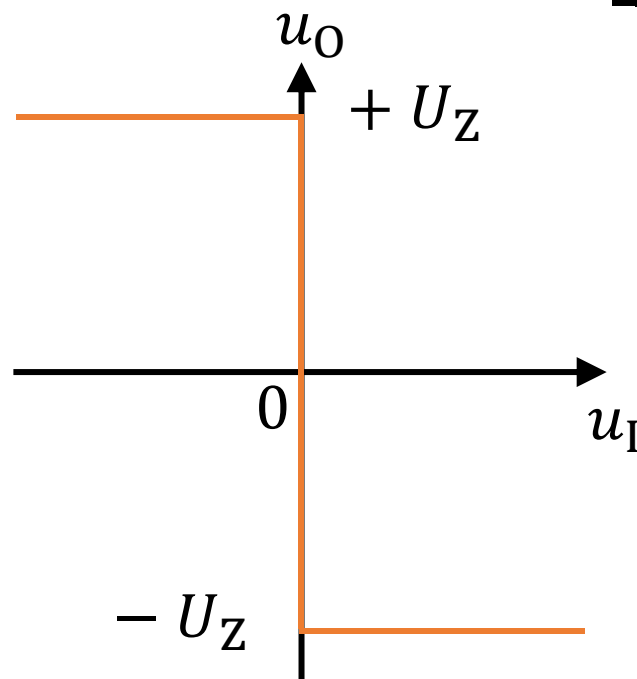
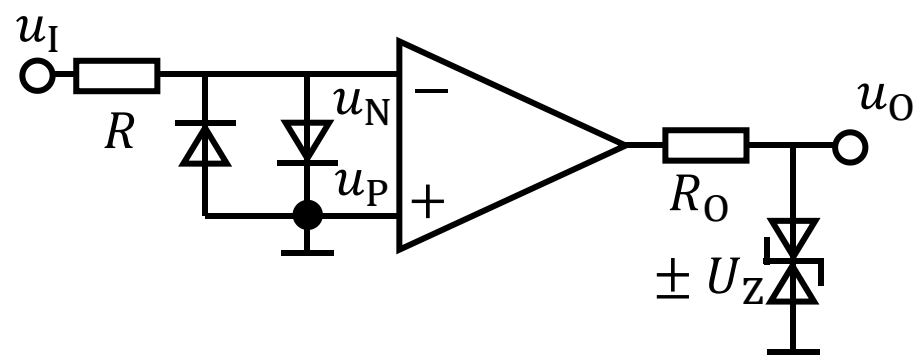
$$\left\{ \begin{array}{ll} u_I > 0 & \Rightarrow u_O = -U_{OM} \\ u_I < 0 & \Rightarrow u_O = +U_{OM} \end{array} \right.$$



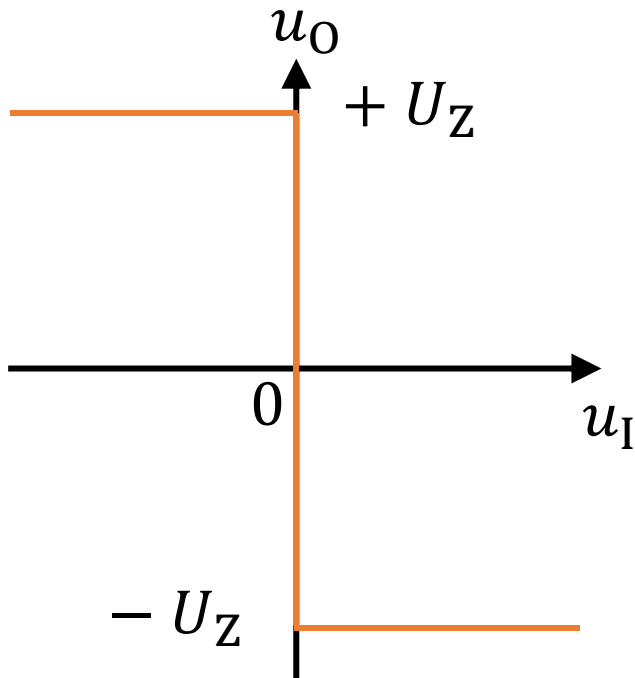




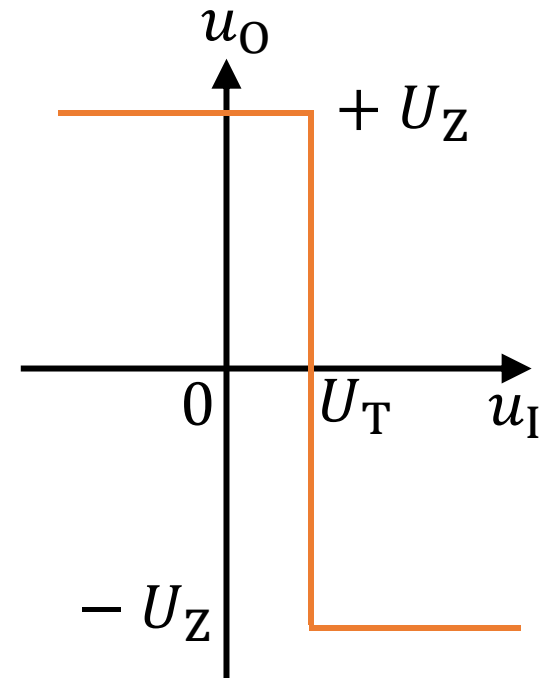
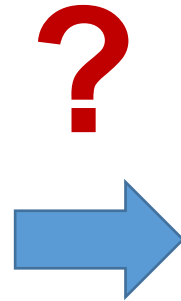
Let $U_Z < U_{OM}$, to achieve desirable output voltage

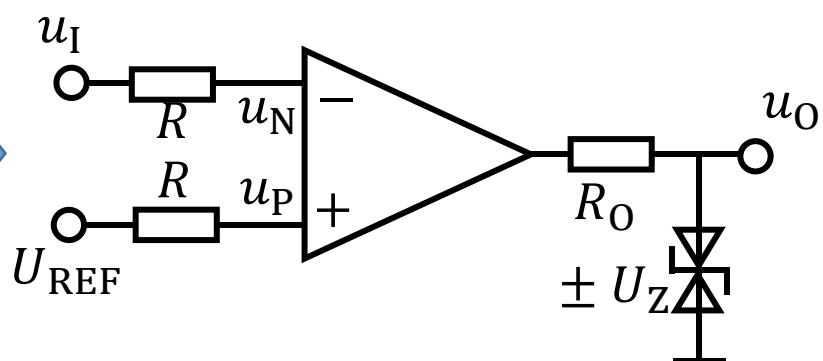
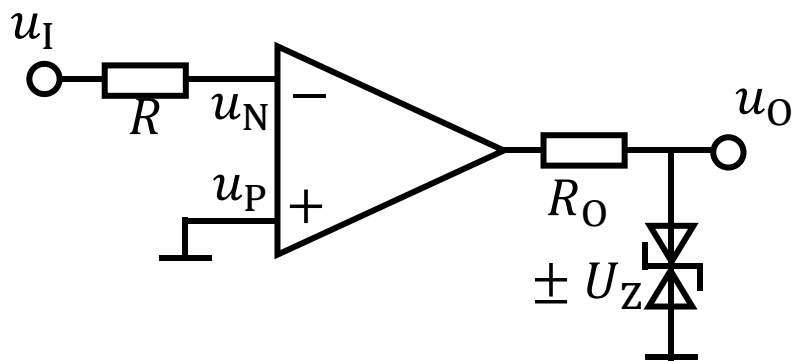
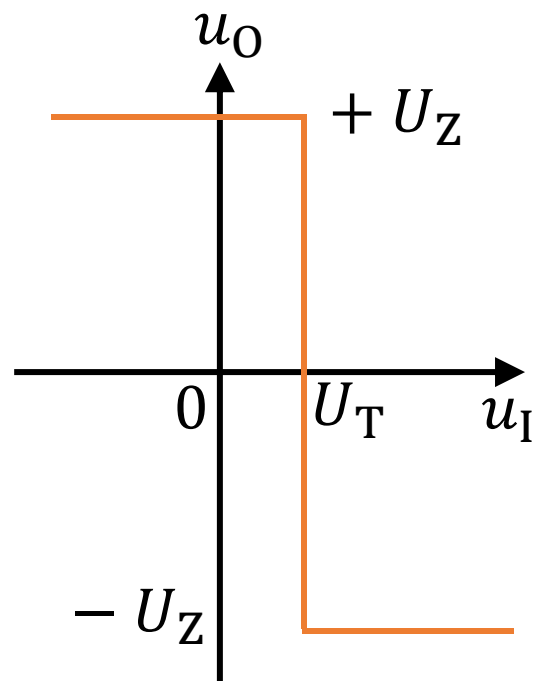
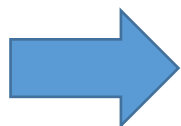
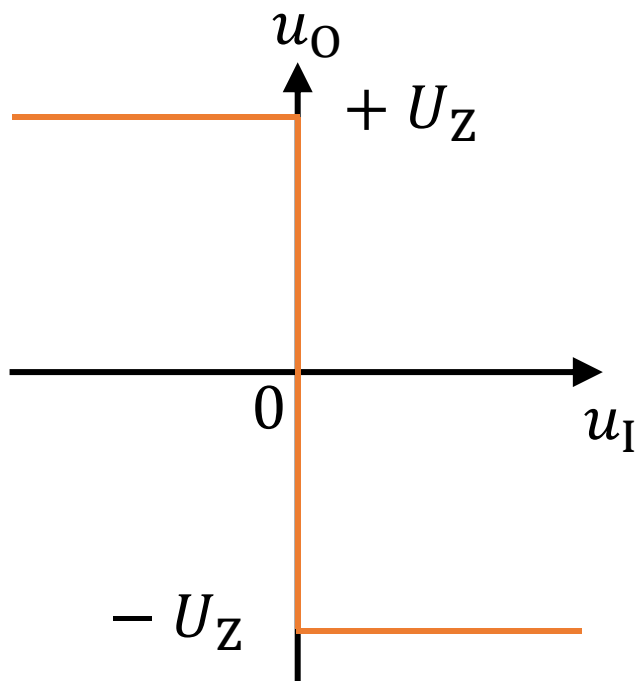


Threshold voltage $U_T = 0$



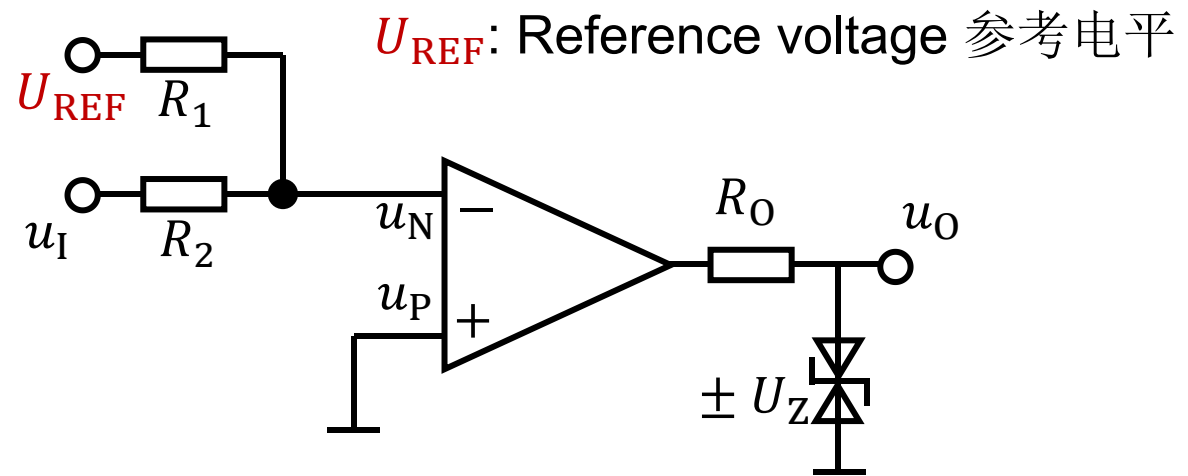
Threshold voltage $U_T \neq 0$





Reference voltage: $U_{REF} = U_T$

(b) Threshold voltage $U_T \neq 0$

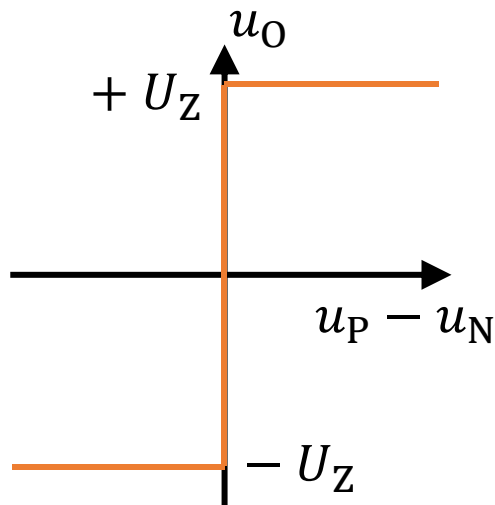


Question 1: Threshold voltage U_T ?

Question 2: $u_O \sim u_I$?

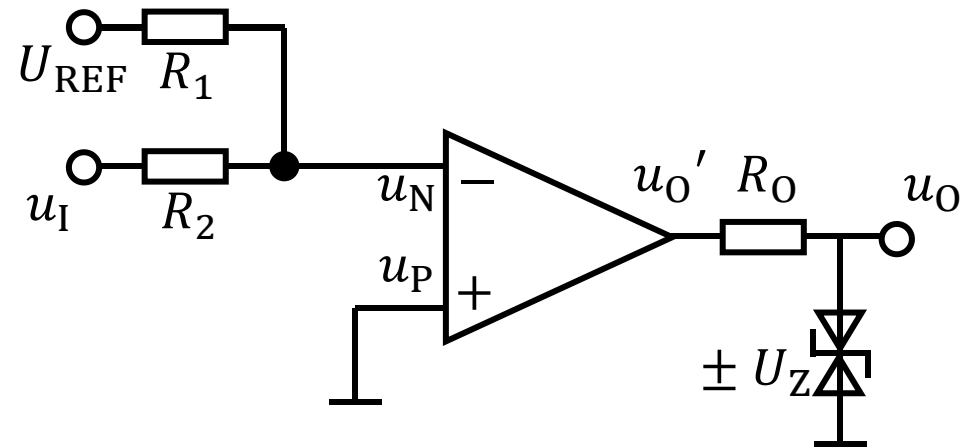
(b) Threshold voltage $U_T \neq 0$

Question 1: U_T ?



$$u_I = f(u_P - u_N)?$$

$$U_T = f(u_P - u_N = 0)?$$



$$u_N = \frac{R_1}{R_1 + R_2} u_I + \frac{R_2}{R_1 + R_2} U_{REF}$$

When $u_P - u_N = 0$, $u_I = U_T$

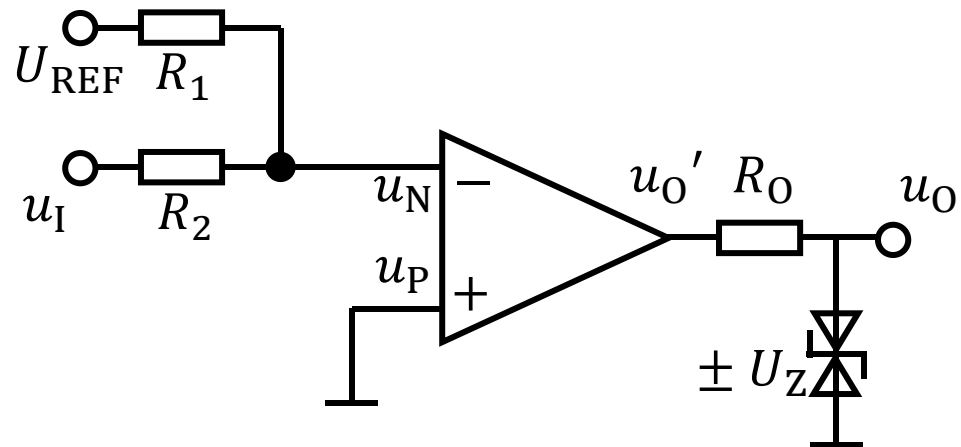


$$0 = \frac{R_1}{R_1 + R_2} U_T + \frac{R_2}{R_1 + R_2} U_{REF}$$

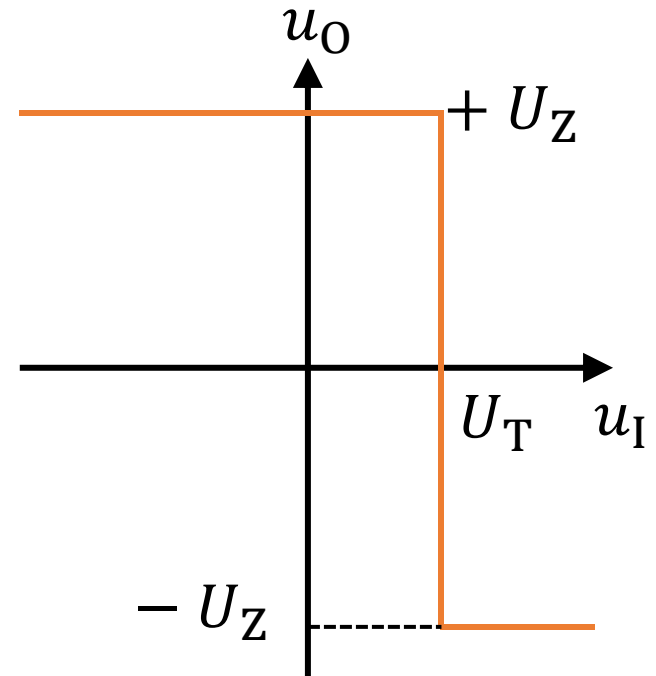


$$U_T = -\frac{R_2}{R_1} U_{REF}$$

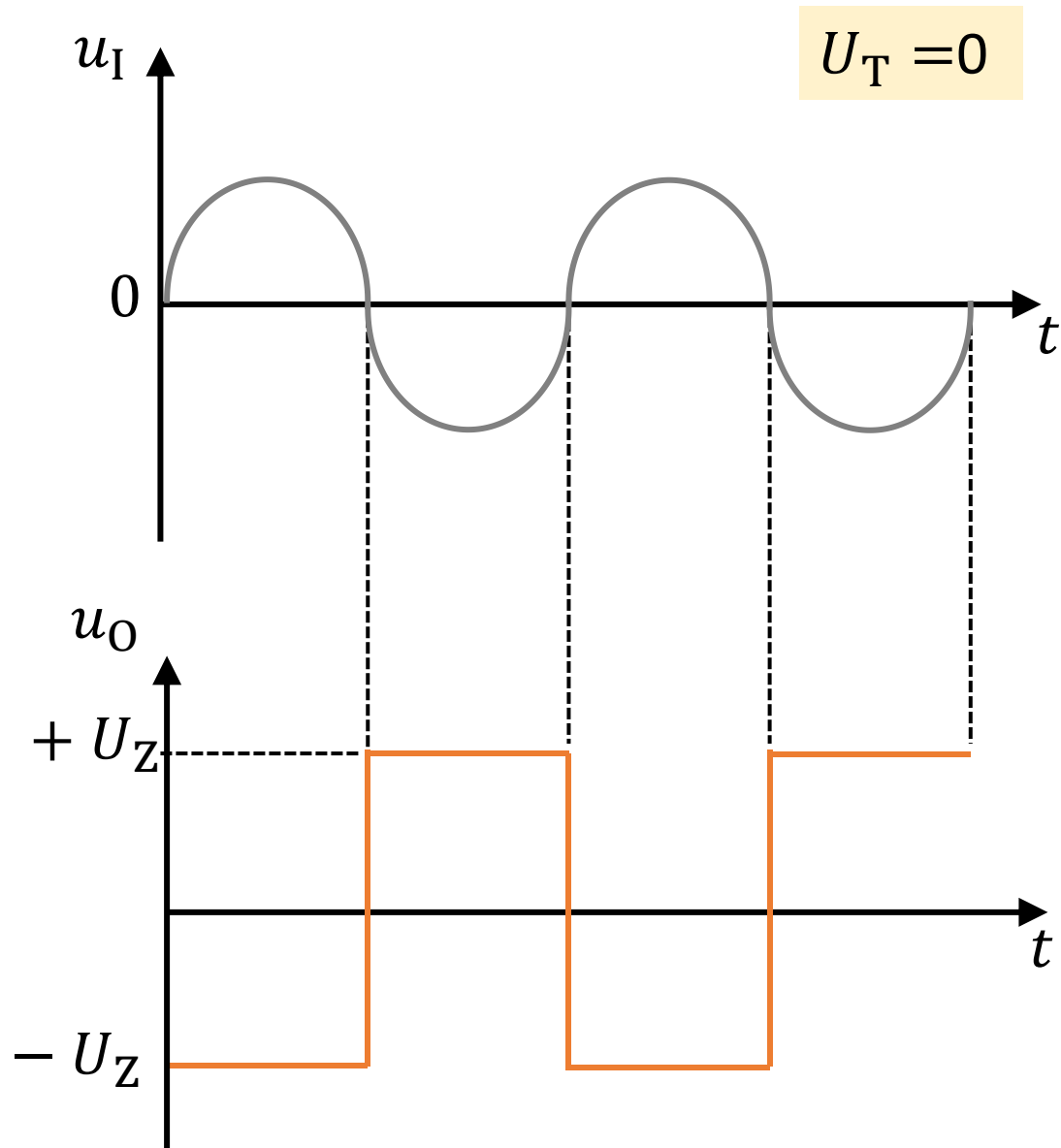
Question 2: $u_O \sim u_I$?



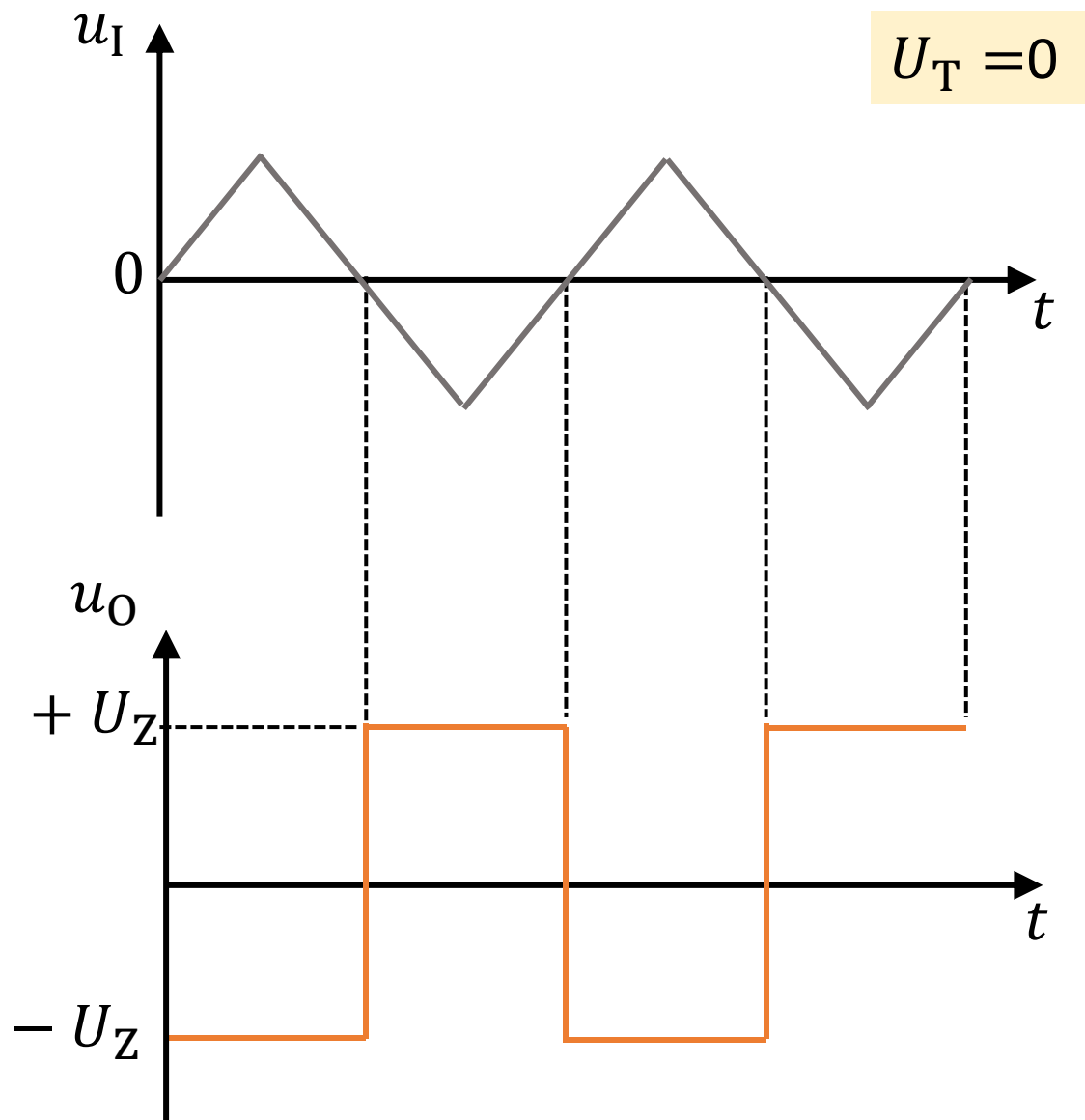
$$U_T = -\frac{R_2}{R_1} U_{REF}$$



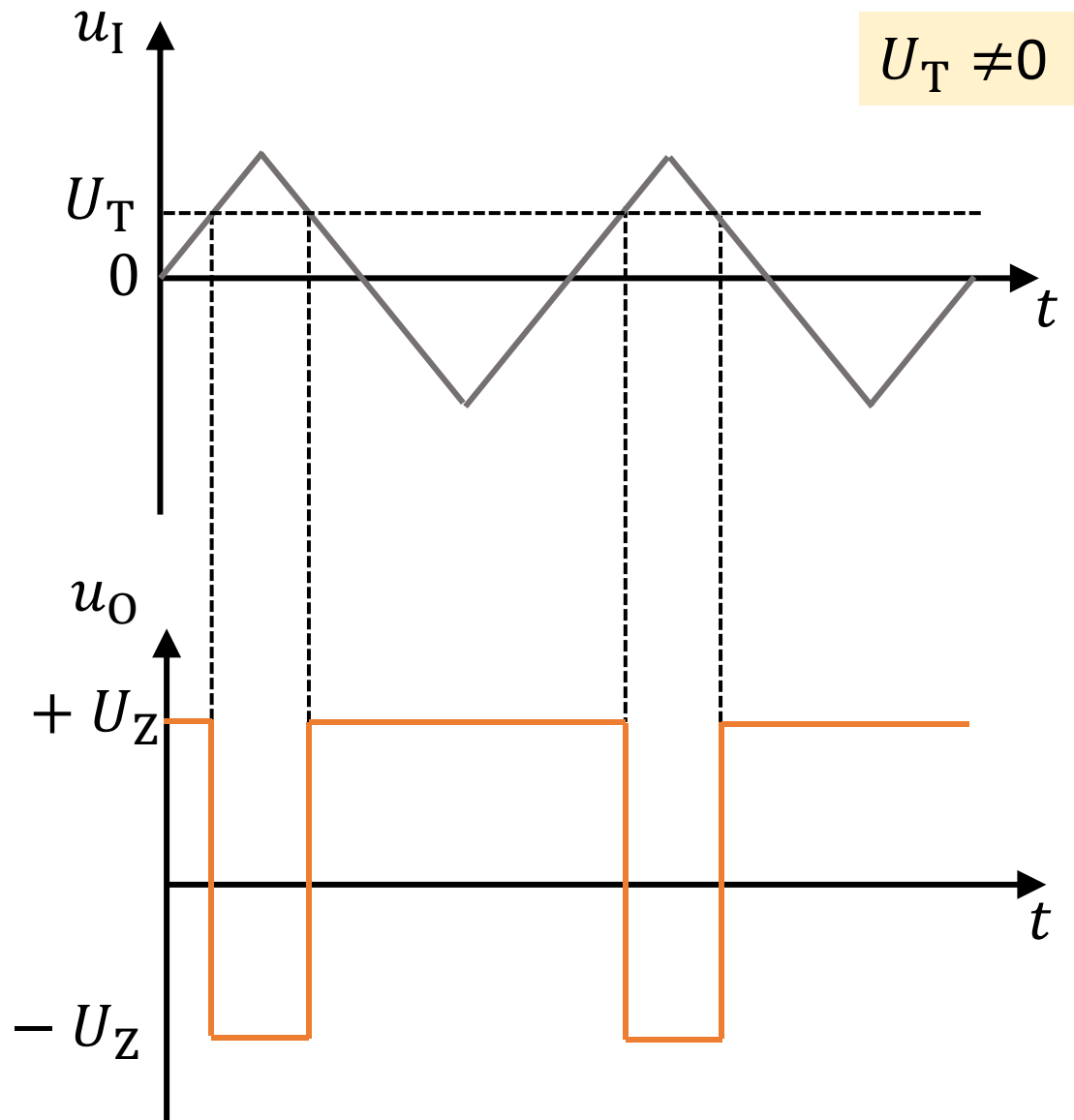
Signal waveform conversion 波形变换



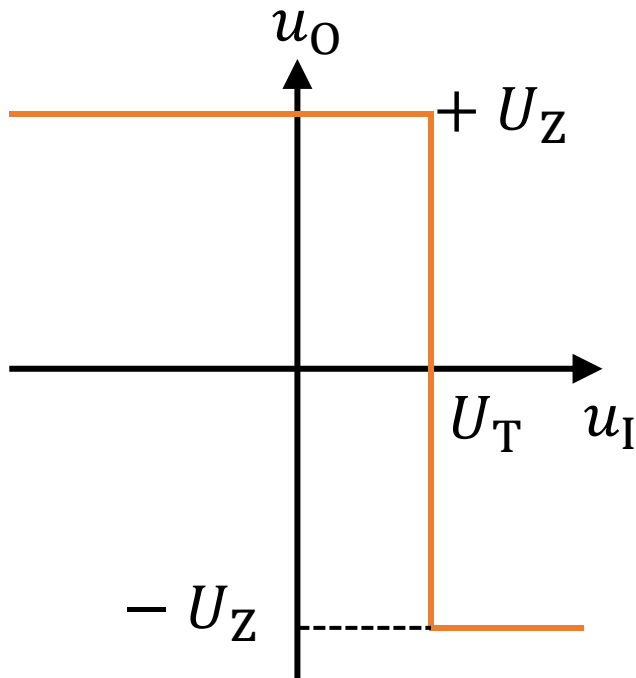
Signal waveform conversion 波形变换



Signal waveform conversion 波形变换



Single-limit voltage comparator



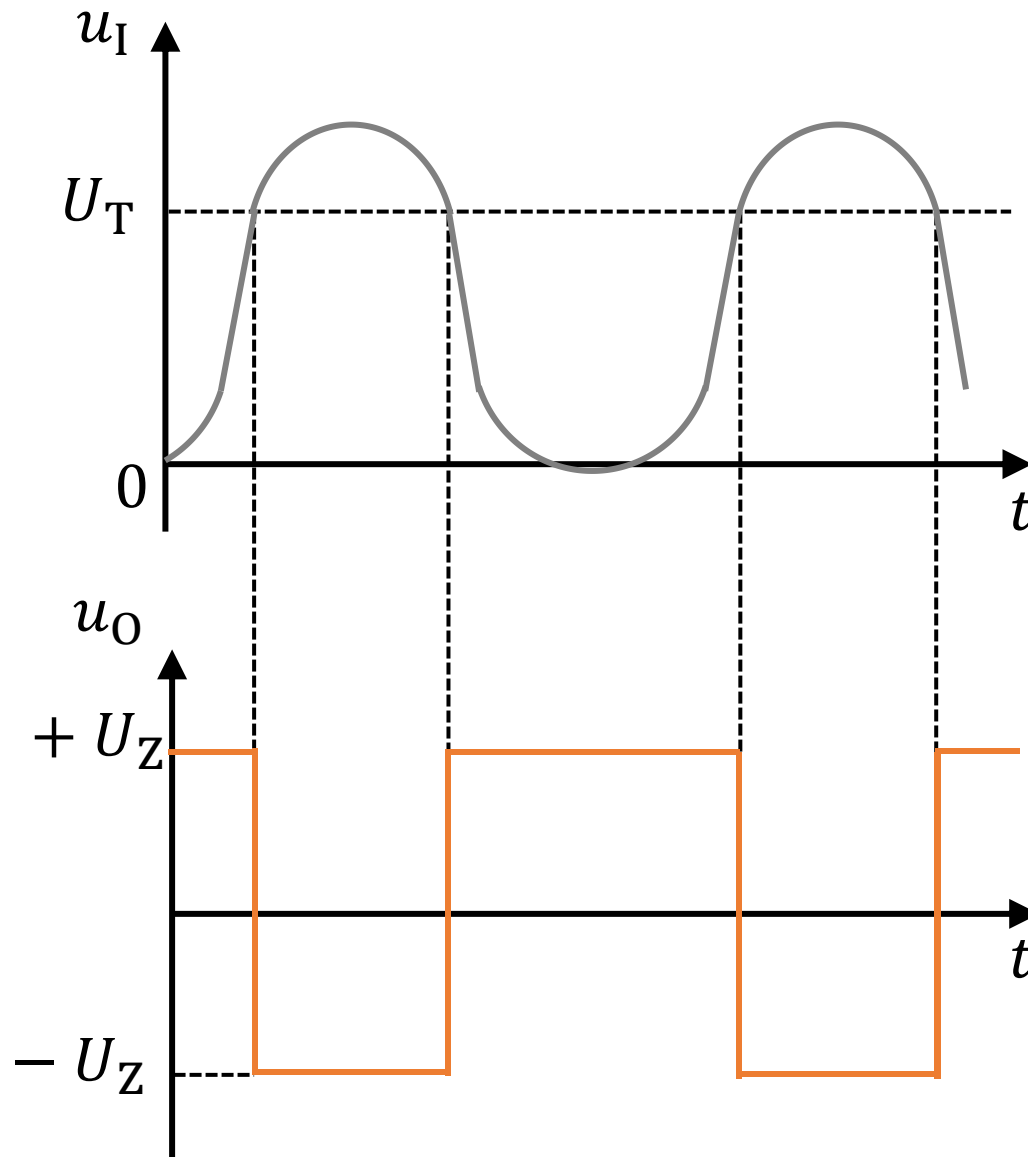
Merits:

- Simple
- High sensitivity 灵敏度高

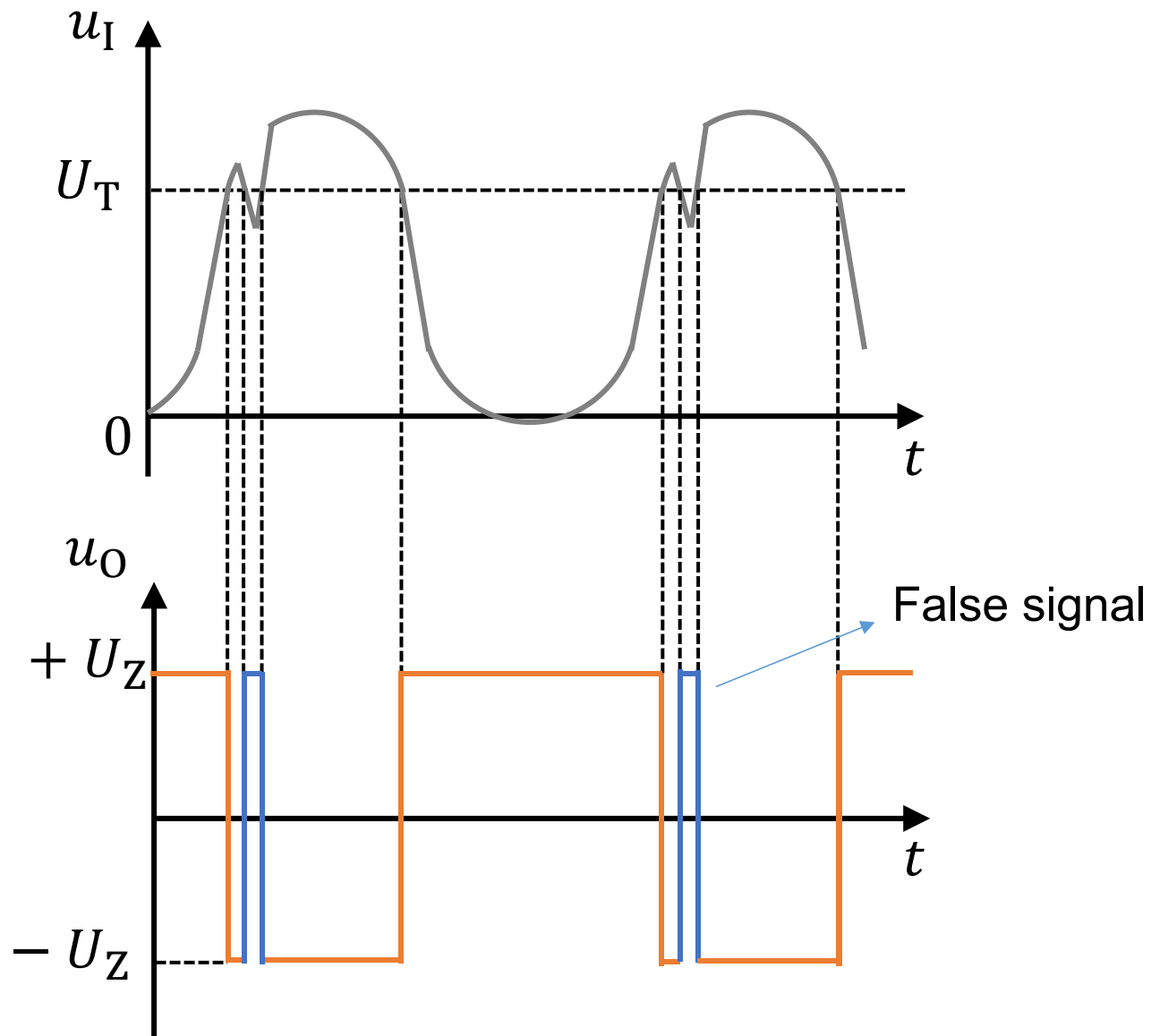
Drawbacks:

Low anti-interference capacity
抗干扰能力弱

Low anti-interference capacity 抗干扰能力弱



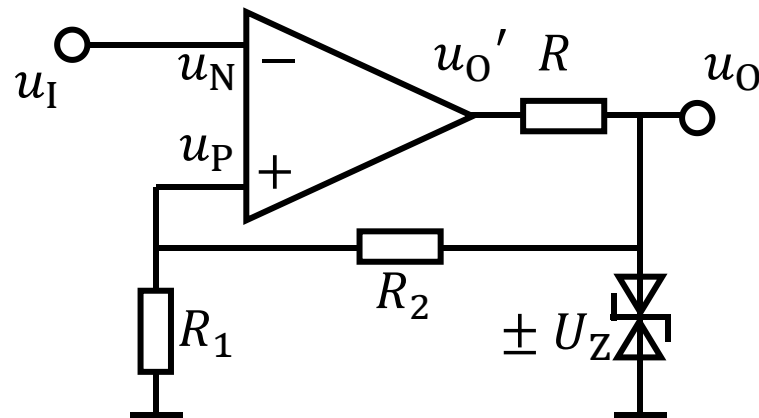
Low anti-interference capacity 抗干扰能力弱



**How to improve the anti-interference
capacity? 如何提高抗干扰能力弱**

(2) Hysteresis comparator 滞回比较器

(a) Inverted input 反相输入



Question1: Positive or negative feedback?

Question2: Linear region or saturation region?

Working principle

$$u_O = \pm U_Z$$

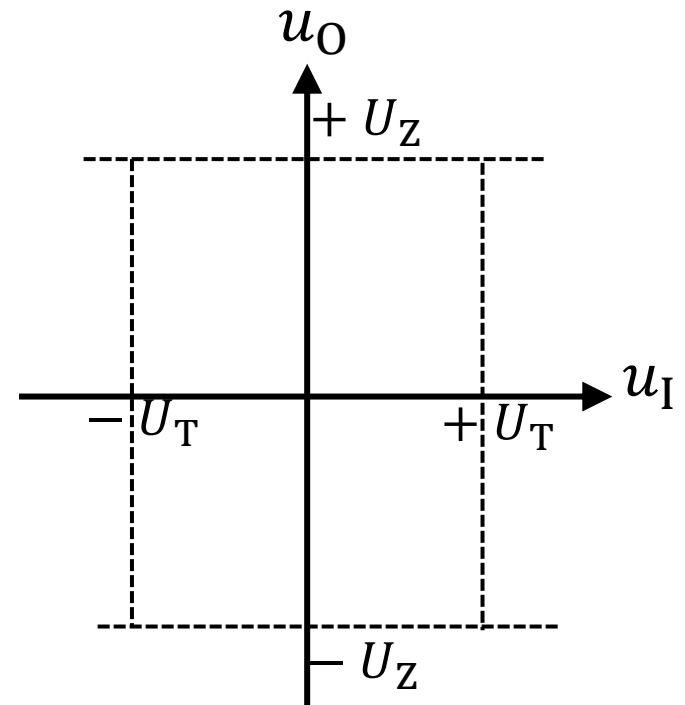
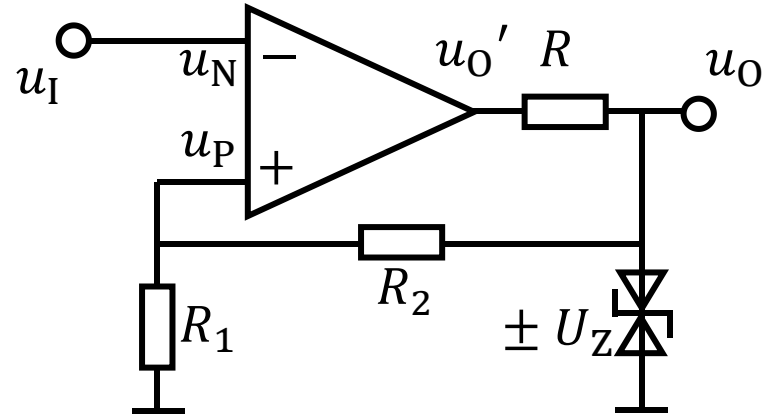
$$u_P = \frac{R_1}{R_1 + R_2} u_O$$

$$u_P - u_N = \frac{R_1}{R_1 + R_2} u_O - u_I$$

Threshold voltage $u_I = U_T$:

$$u_P - u_N = \frac{R_1}{R_1 + R_2} u_O - U_T = 0$$

$$U_T = \frac{R_1}{R_1 + R_2} u_O = \pm \frac{R_1}{R_1 + R_2} U_Z$$



Working principle

(i) u_I : from $-U_0$ to $+U_0$

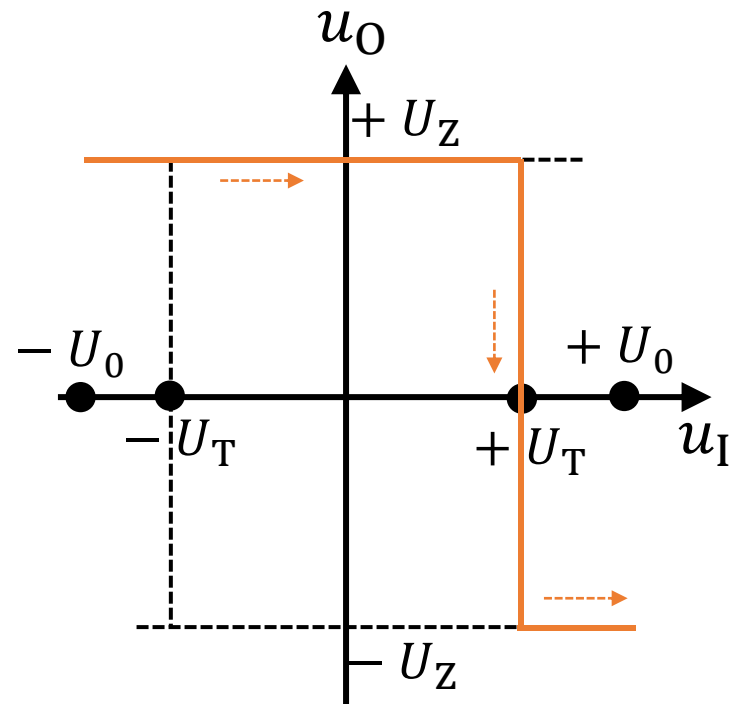
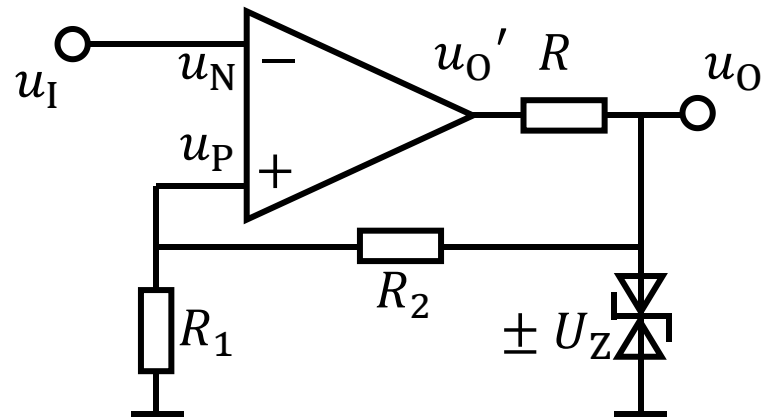
At starting point: $u_I = -U_0$

$$u_O = +U_Z$$

$$+U_T = +\frac{R_1}{R_1 + R_2} U_Z$$

After u_I crosses $+U_T$

$$u_O = -U_Z$$



Working principle

(ii) u_I : from $+U_0$ to $-U_0$

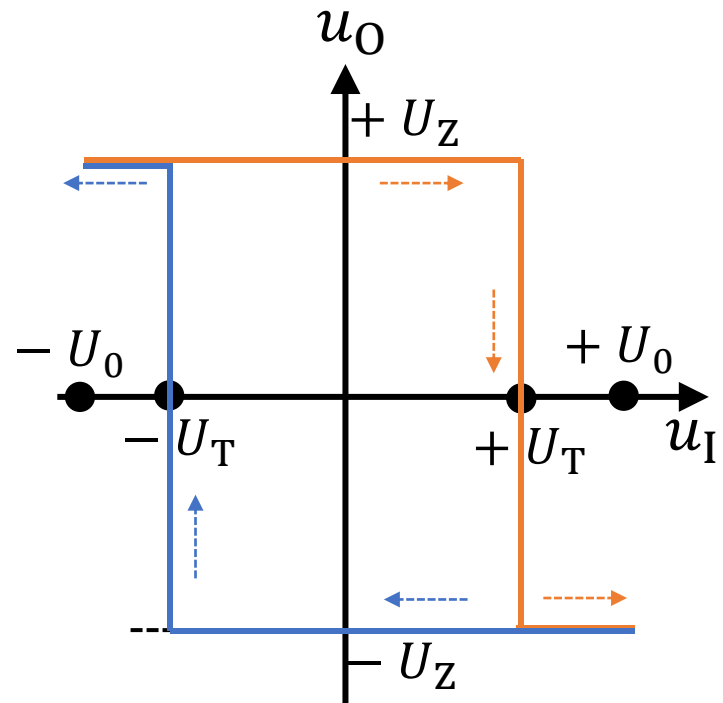
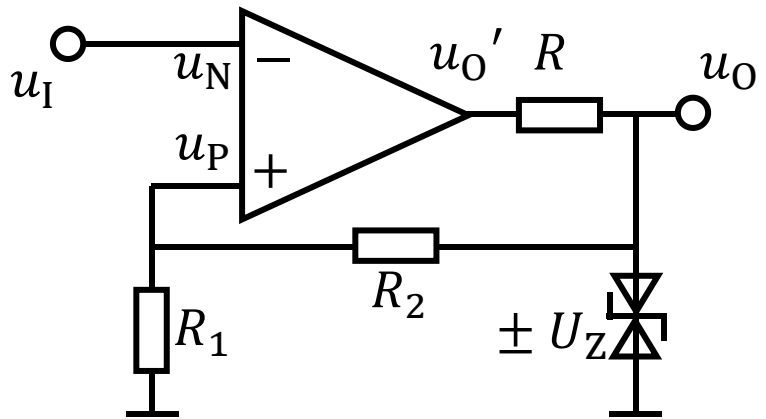
At starting point: $u_I = +U_0$

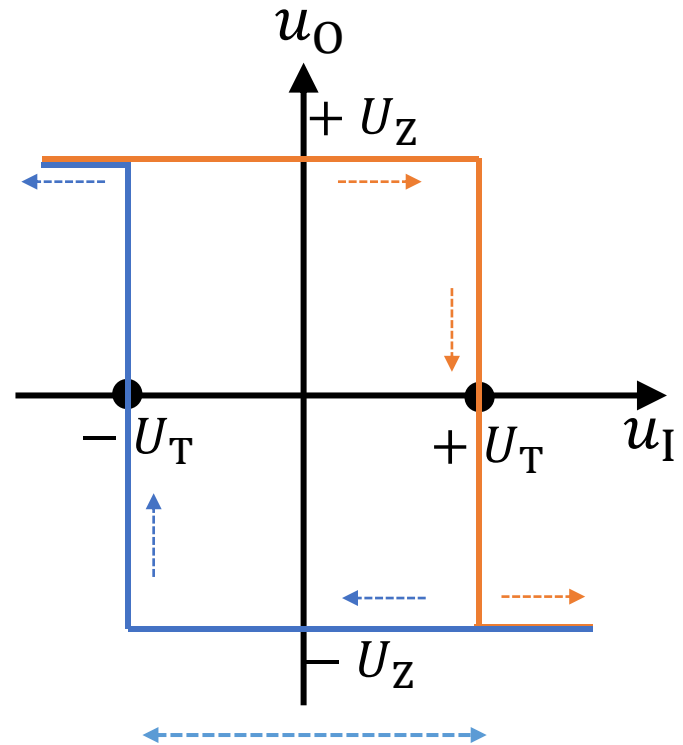
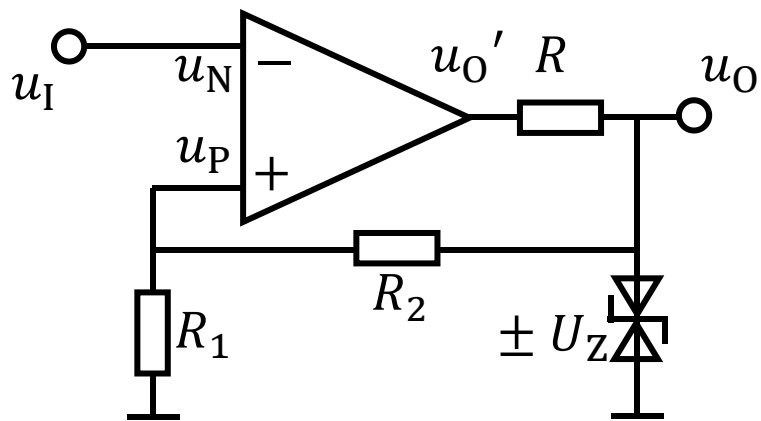
$$u_O = -U_Z$$

$$-U_T = -\frac{R_1}{R_1 + R_2} U_Z$$

After u_I crosses $-U_T$

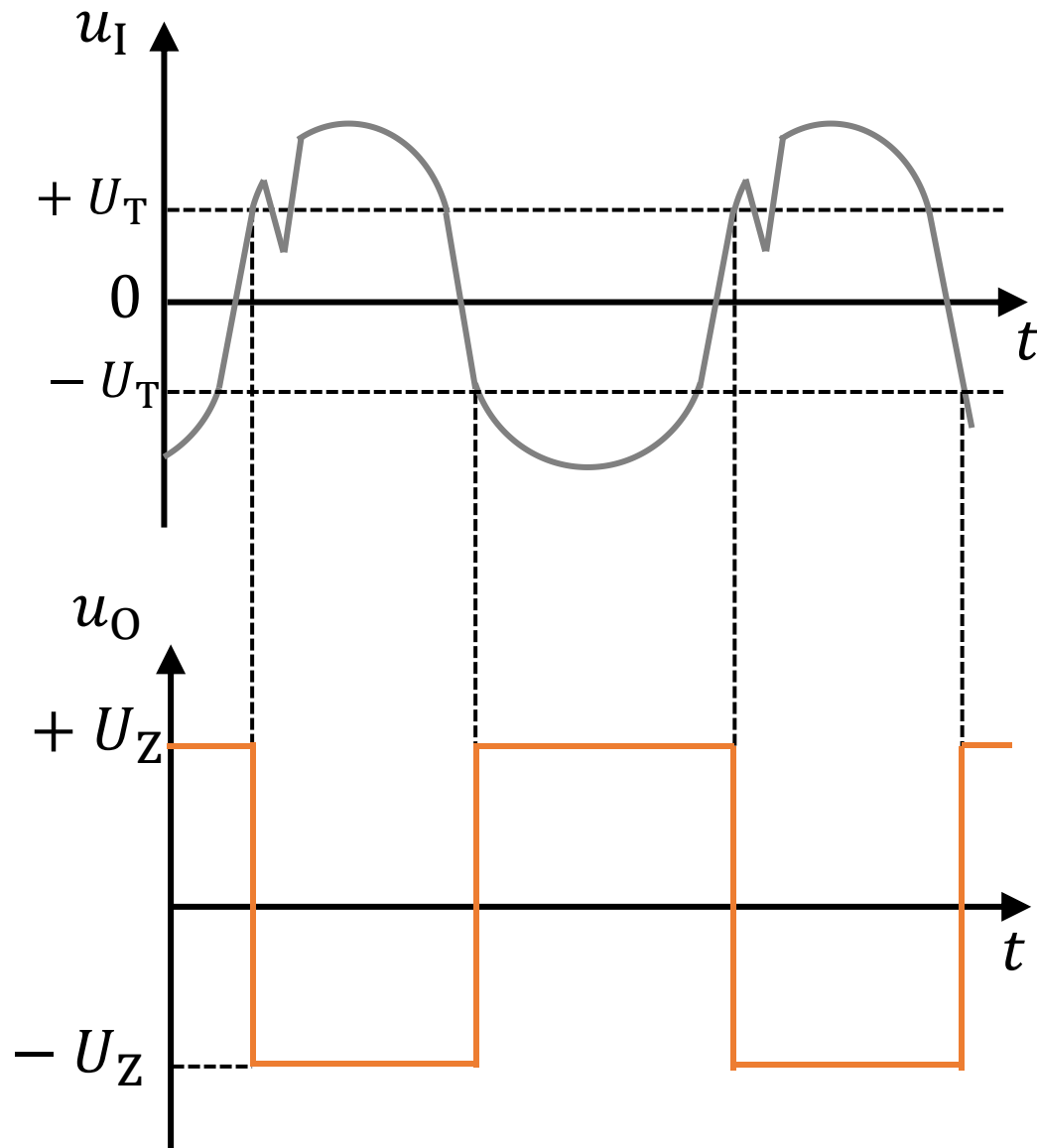
$$u_O = +U_Z$$





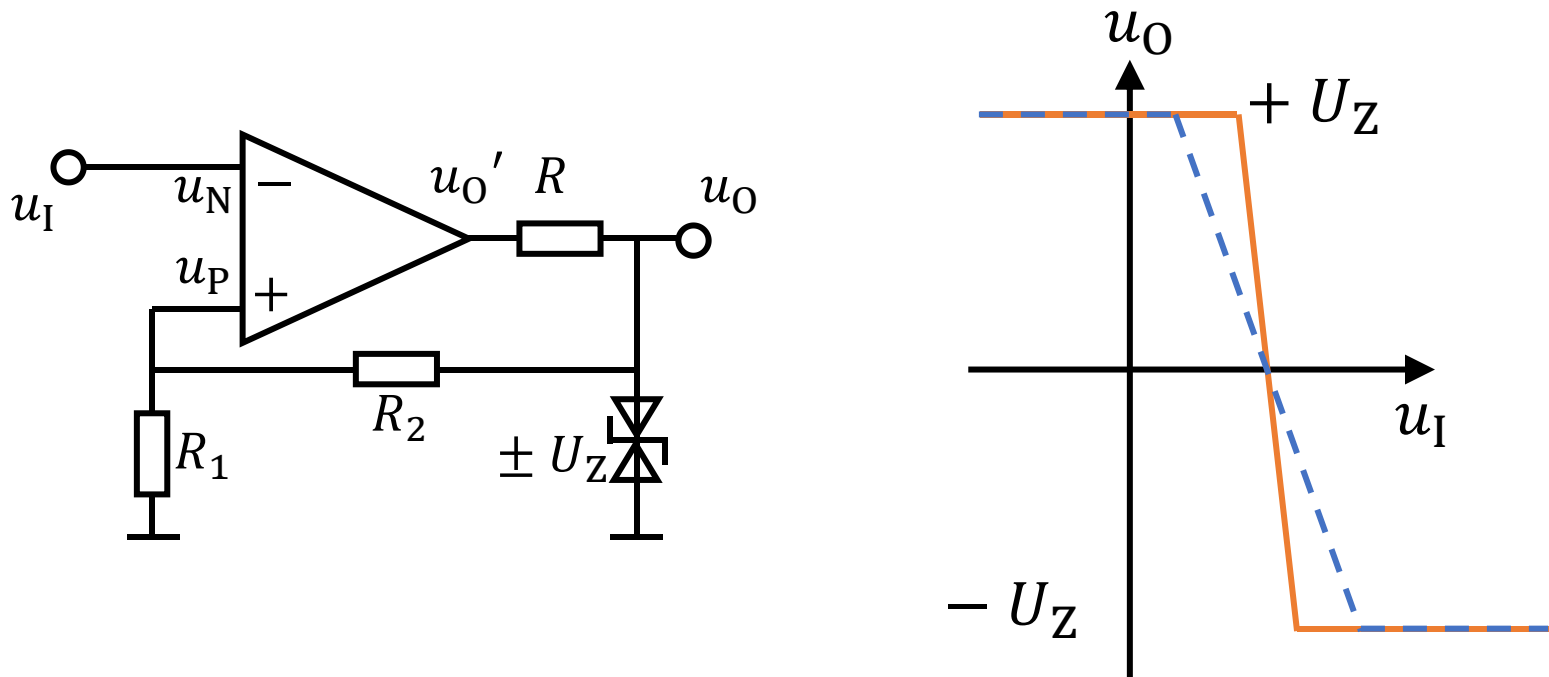
Cannot distinguish signal variation smaller than $2U_T$

High anti-interference capacity 抗干扰能力强



Positive feedback: can accelerate the transit process of u_O : $-U_Z \leftrightarrow +U_Z$

正反馈：可以加速输出电压转变速度 $-U_Z \leftrightarrow +U_Z$



Positive feedback can increase $|u_P - u_N|$

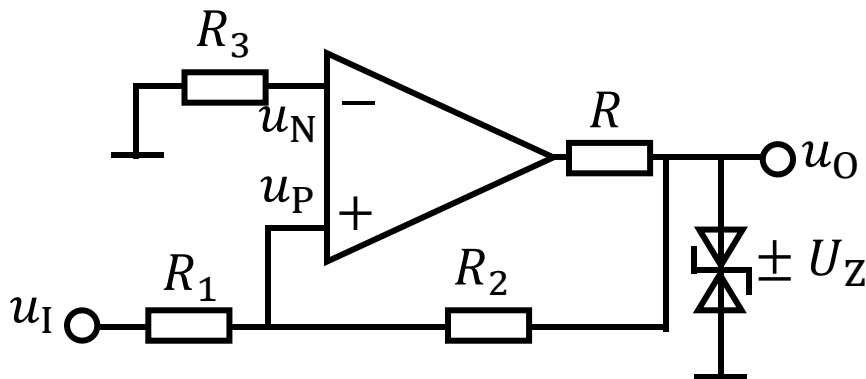
(b) Non-inverted input 同相输入

Working principle

$$u_O = \pm U_Z$$

$$u_P = \frac{R_1}{R_1 + R_2} u_O + \frac{R_2}{R_1 + R_2} u_I$$

$$u_P - u_N = \frac{R_1}{R_1 + R_2} u_O + \frac{R_2}{R_1 + R_2} u_I$$



Threshold voltage $u_I = U_T$:

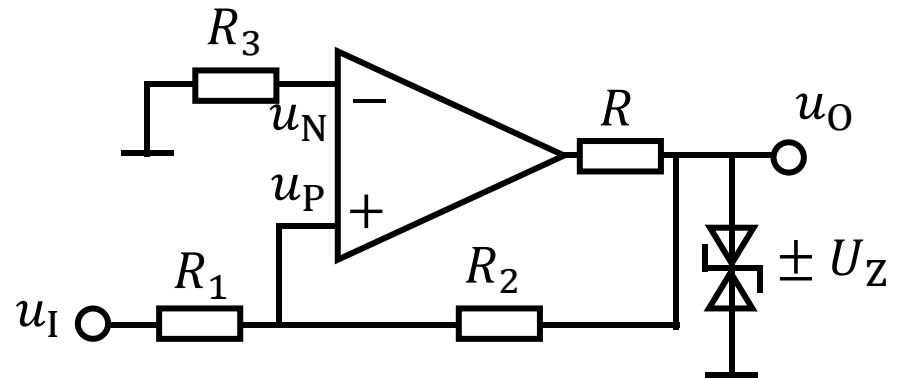
$$u_P - u_N = \frac{R_1}{R_1 + R_2} u_O + \frac{R_2}{R_1 + R_2} U_T = 0$$

$$U_T = -\frac{R_1}{R_2} u_O = \mp \frac{R_1}{R_2} U_Z$$

Working principle

(i) u_I : from $-U_0$ to $+U_0$

At starting point: $u_I = -U_0$

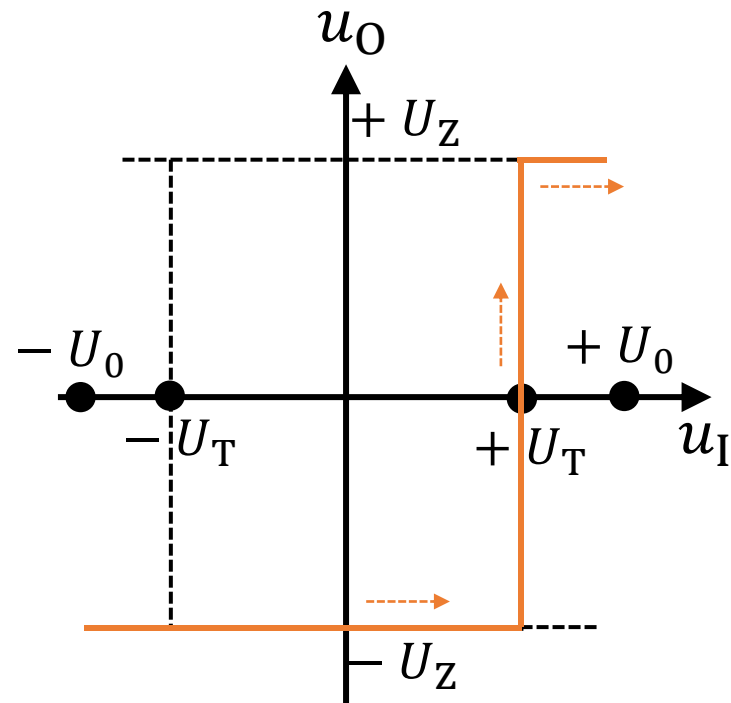


$$u_O = -U_Z$$

$$+U_T = +\frac{R_1}{R_2}U_Z$$

After u_I crosses $+U_T$

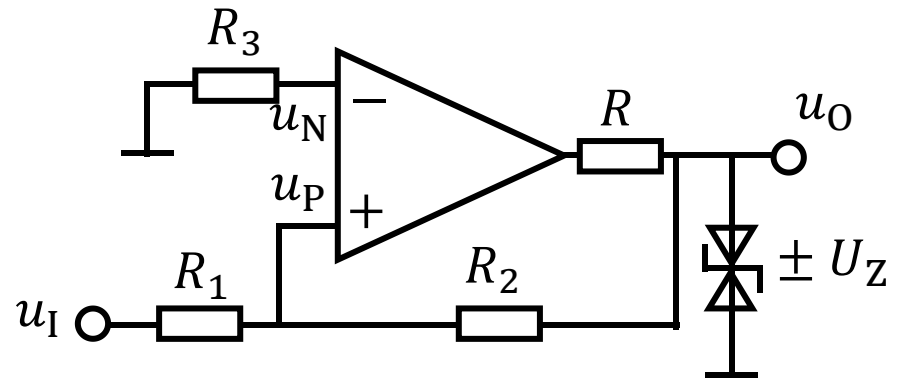
$$u_O = +U_Z$$



Working principle

(ii) u_I : from $+U_0$ to $-U_0$

At starting point: $u_I = +U_0$

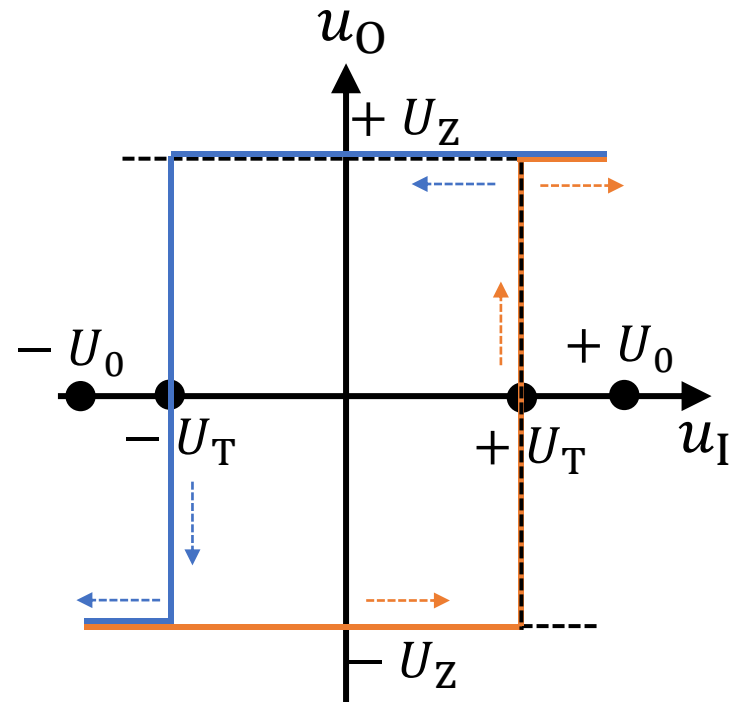


$$u_O = +U_Z$$

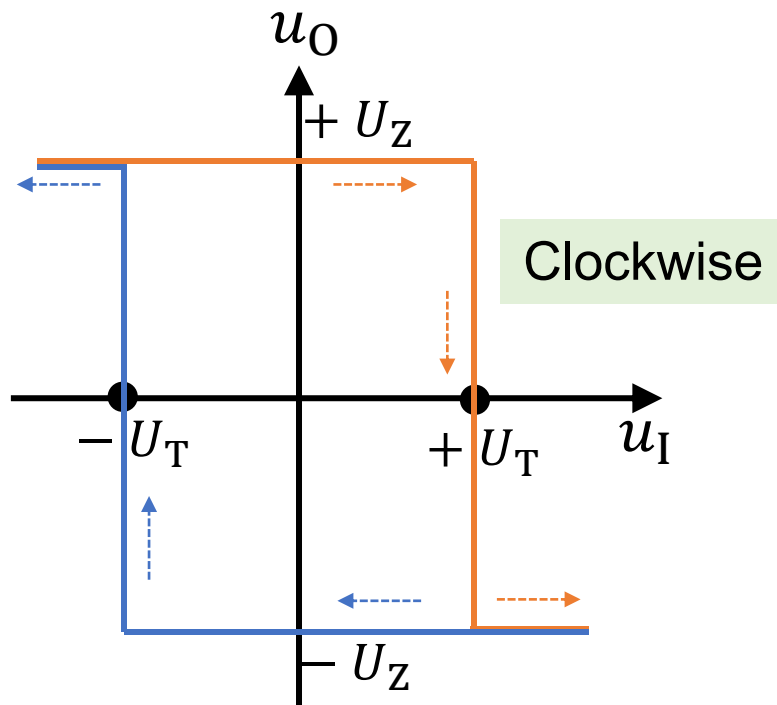
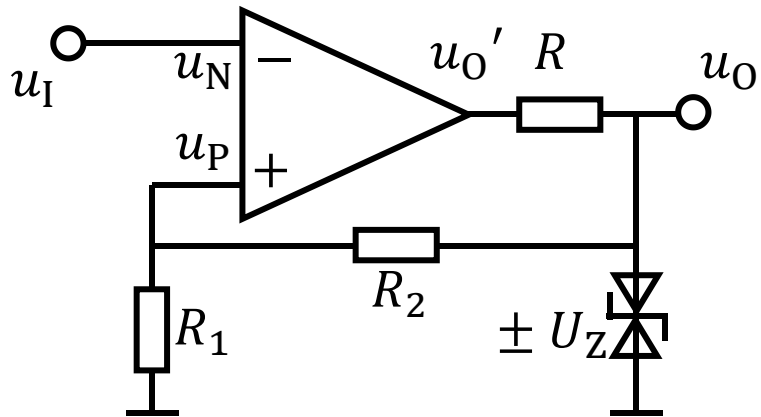
$$-U_T = -\frac{R_1}{R_2} U_Z$$

After u_I crosses $-U_T$

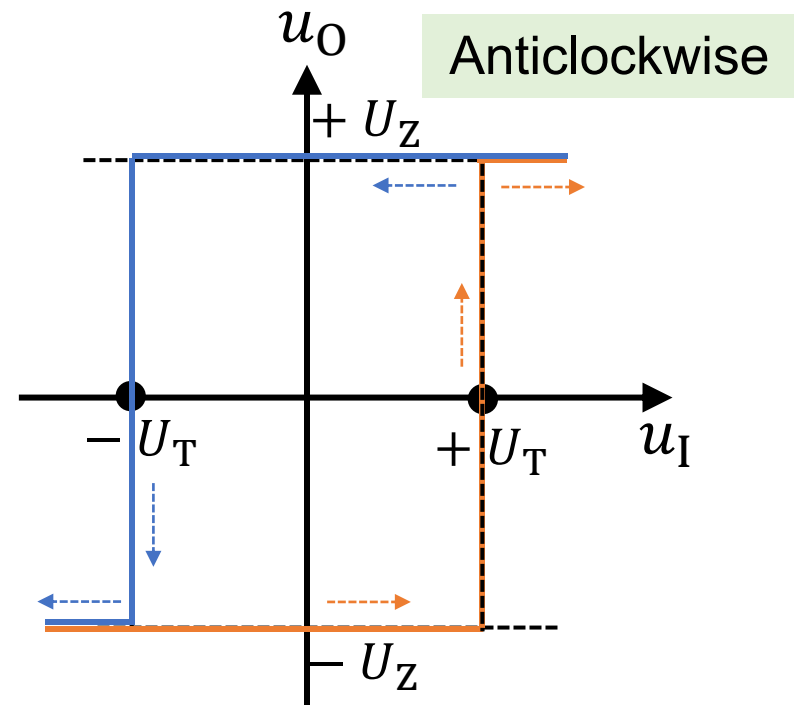
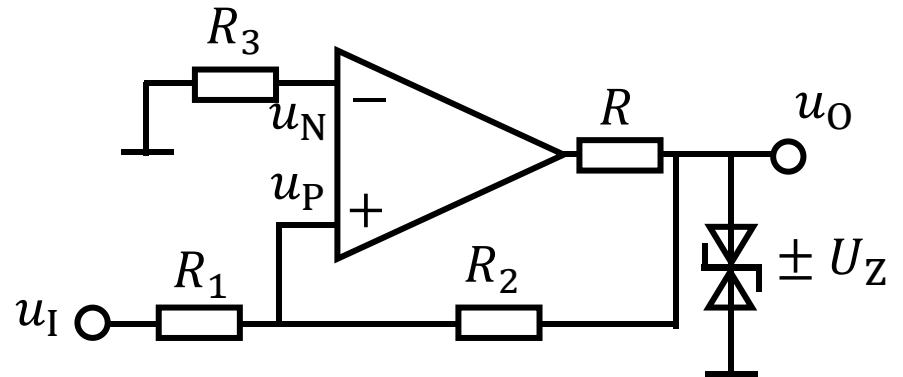
$$u_O = -U_Z$$



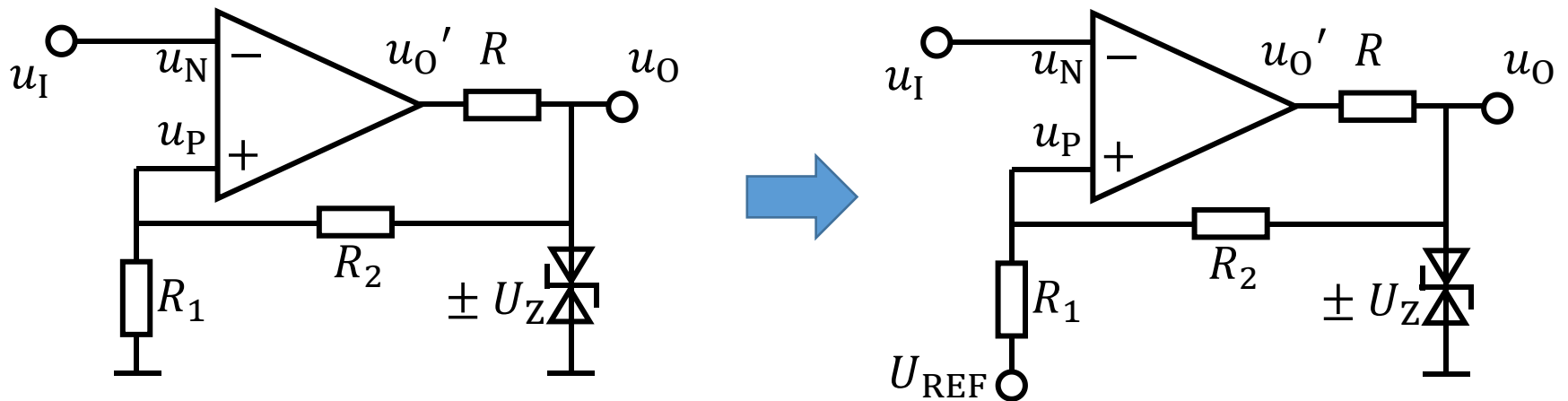
Inverted input



Non-inverted input



(c) Hysteresis comparator with reference voltage



Working principle: $u_I = U_T$?

$$u_O = \pm U_Z$$

$$u_P = \frac{R_1}{R_1 + R_2} u_O + \frac{R_2}{R_1 + R_2} U_{REF}$$

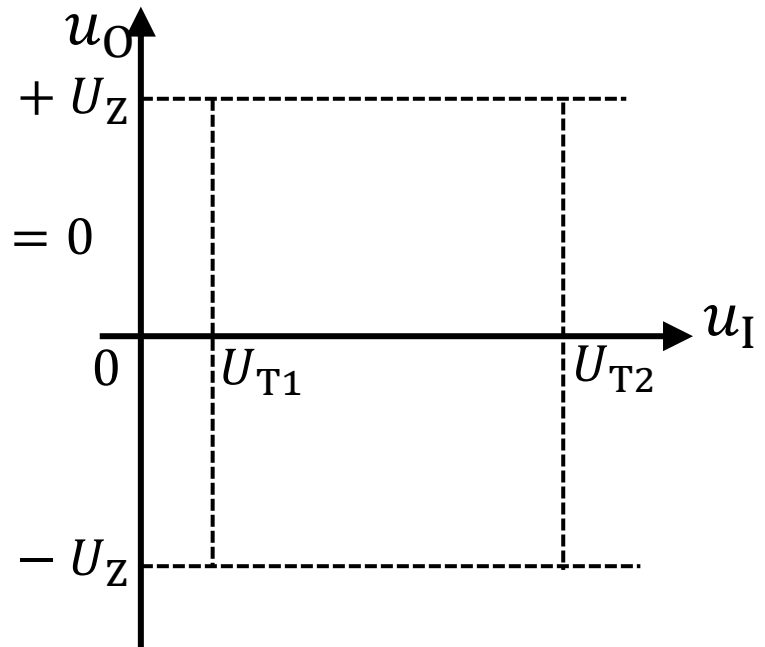
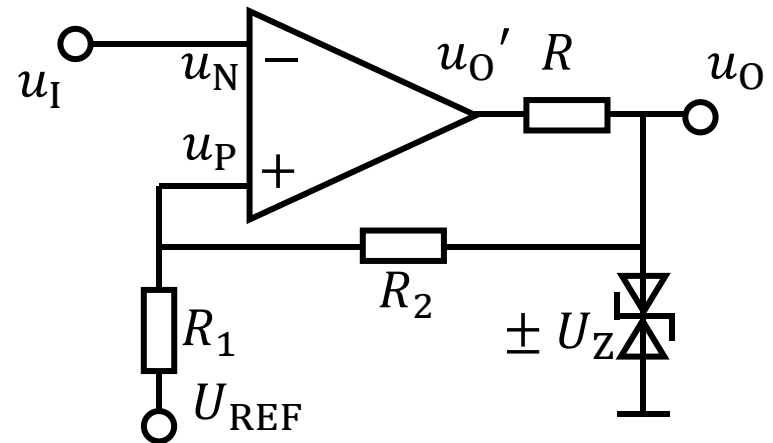
$$\begin{aligned} u_P - u_N \\ = \frac{R_1}{R_1 + R_2} u_O + \frac{R_2}{R_1 + R_2} U_{REF} - u_I \end{aligned}$$

Threshold voltage $u_I = U_T$:

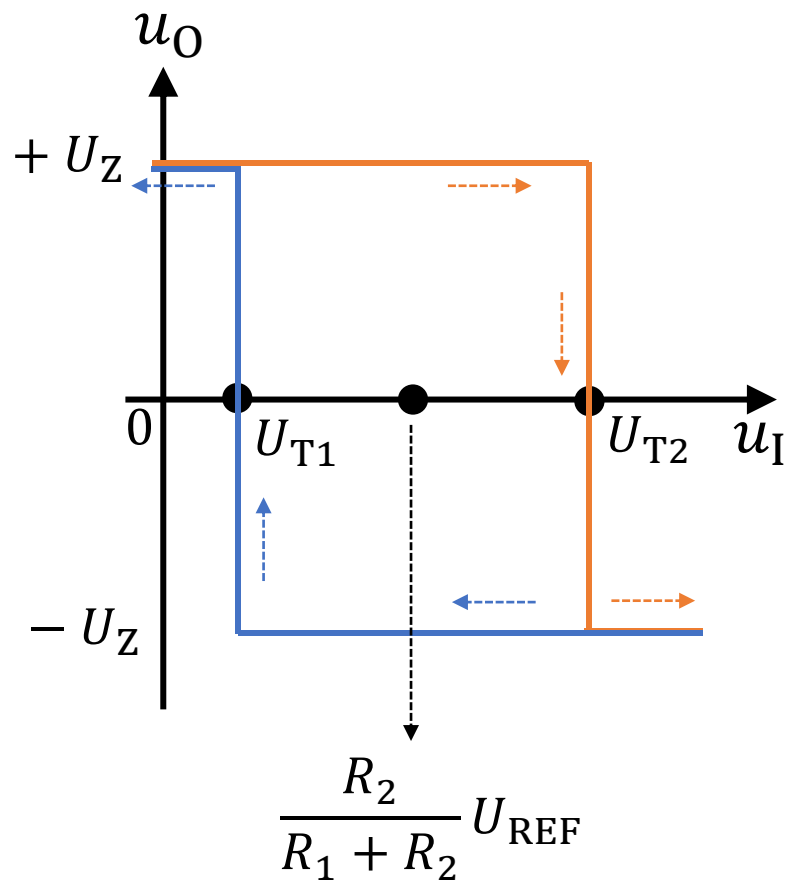
$$u_P - u_N = \frac{R_1}{R_1 + R_2} u_O + \frac{R_2}{R_1 + R_2} U_{REF} - U_T = 0$$

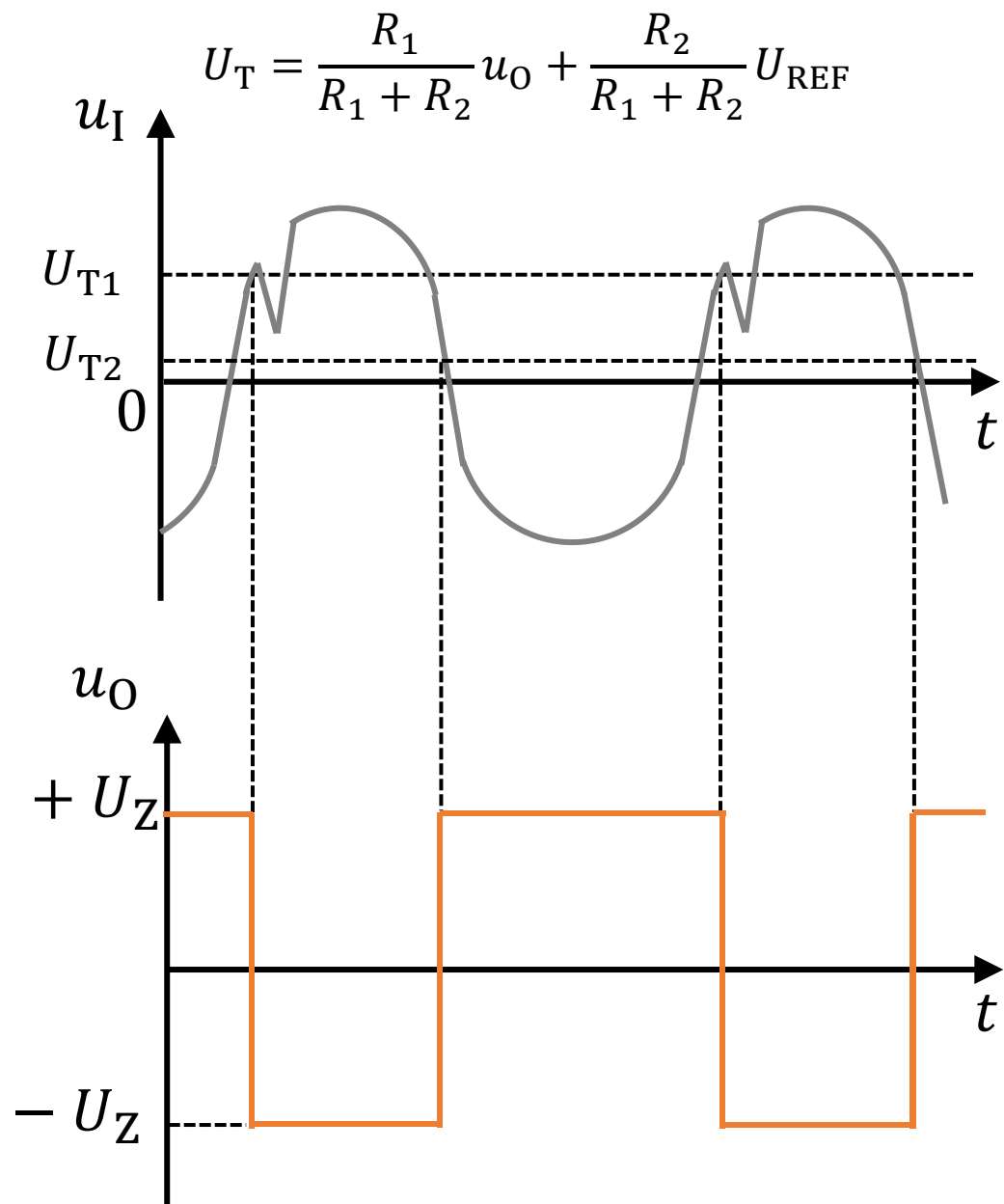
$$U_T = \frac{R_1}{R_1 + R_2} u_O + \frac{R_2}{R_1 + R_2} U_{REF}$$

$$= \pm \frac{R_1}{R_1 + R_2} U_Z + \frac{R_2}{R_1 + R_2} U_{REF}$$

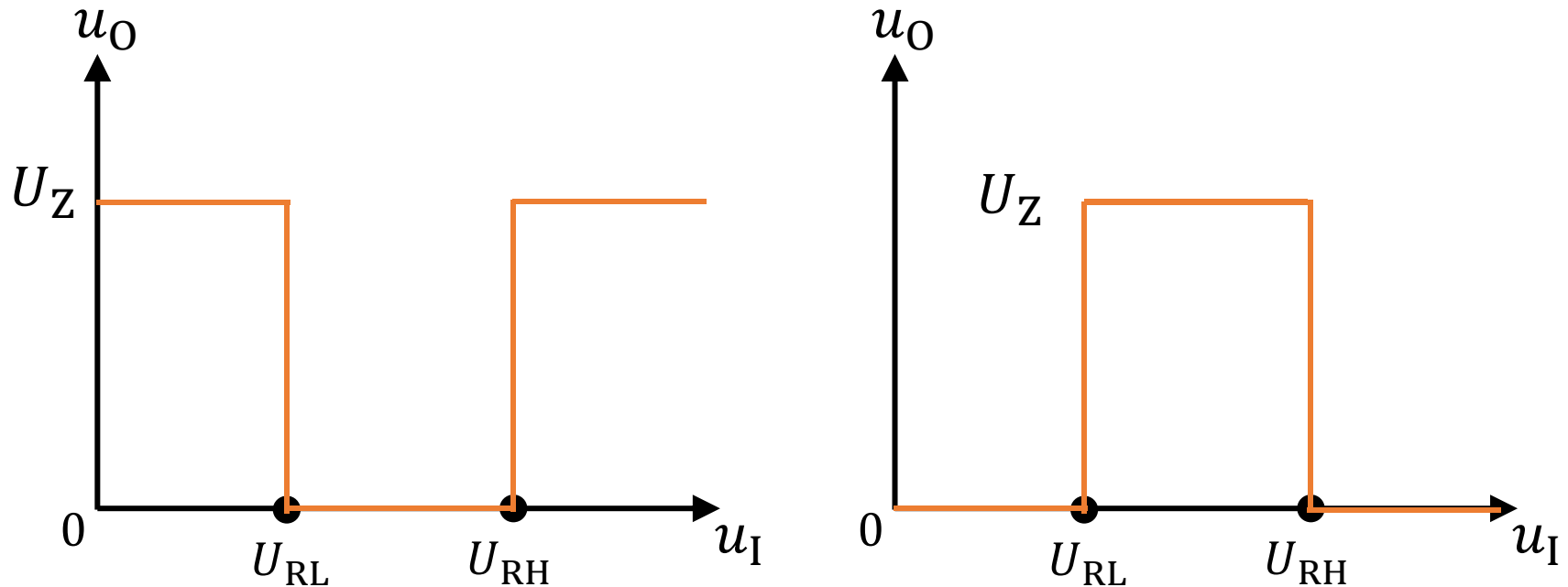


$$U_T = \frac{R_1}{R_1 + R_2} u_O + \frac{R_2}{R_1 + R_2} U_{\text{REF}}$$



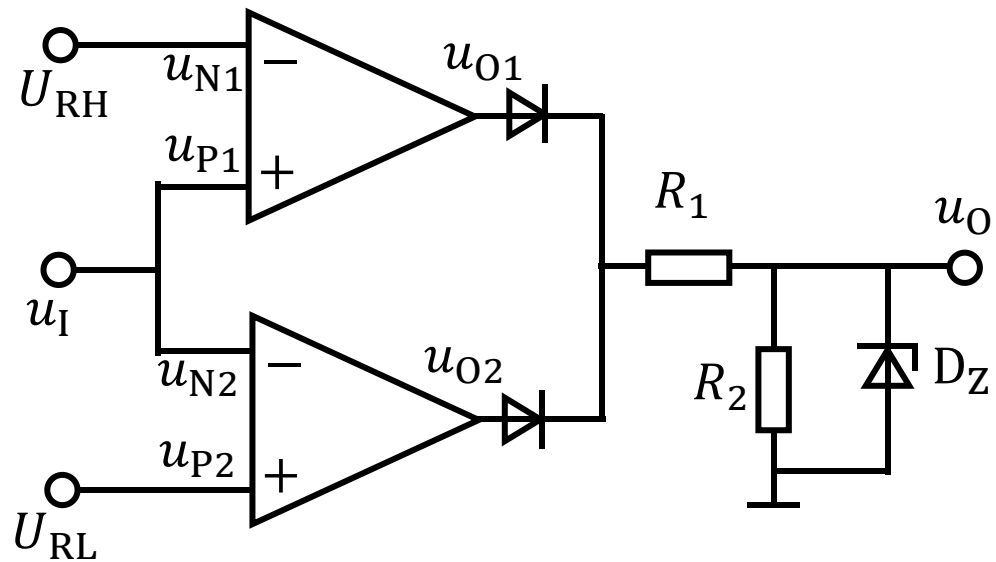


(3) Window comparator 窗口比较器



Can be applied in industrial control systems: measuring the range of temperature, pressure or liquid level. 用于工业控制系统, 测量温度、压力、液面等的范围。

(3) Window comparator 窗口比较器



Working principle

$$U_{RH} > U_{RL}$$

(i) When $u_I > U_{RH}$

$$u_{O1} = +U_{OM} \quad \text{D1 is on}$$

$$u_{O2} = -U_{OM} \quad \text{D2 is off}$$

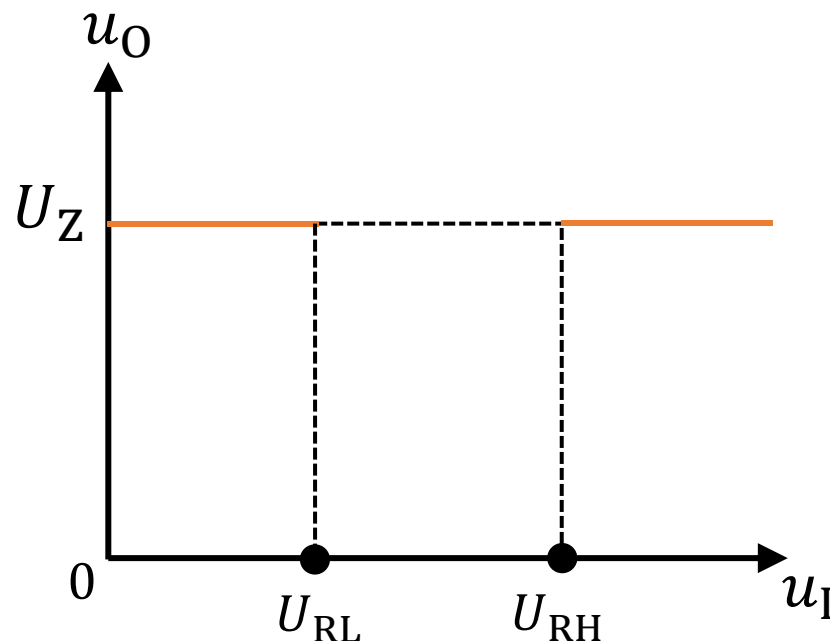
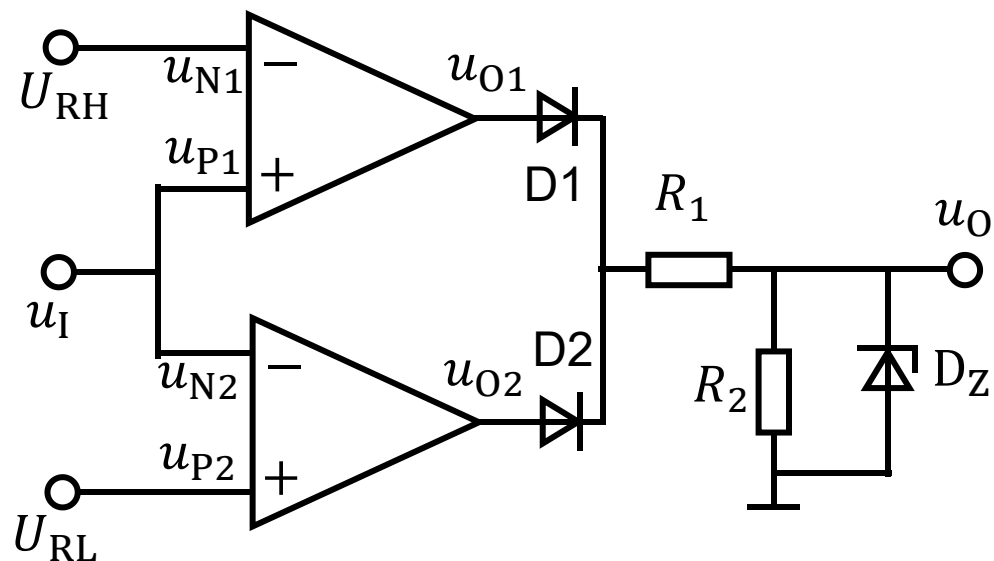
$$u_O = U_Z$$

(ii) When $u_I < U_{RL}$

$$u_{O1} = -U_{OM} \quad \text{D1 is off}$$

$$u_{O2} = +U_{OM} \quad \text{D2 is on}$$

$$u_O = U_Z$$



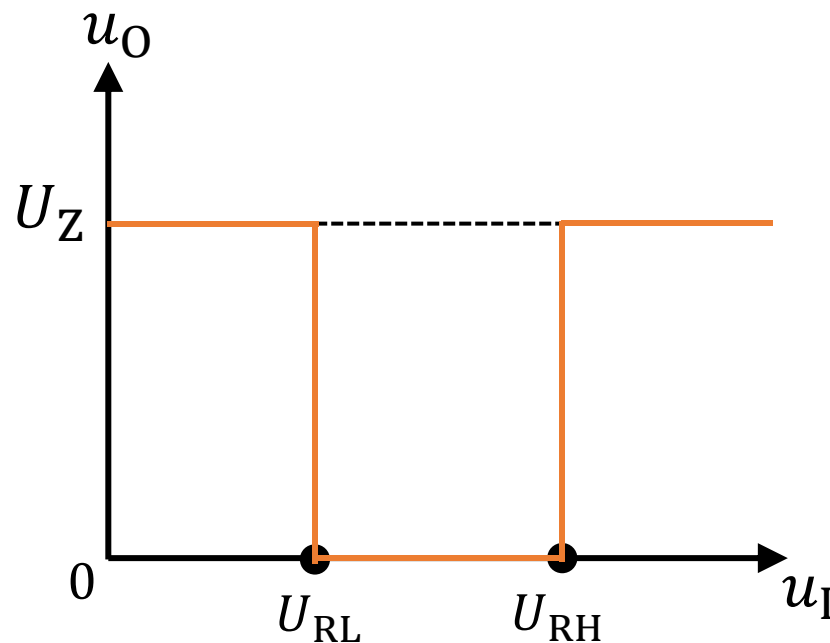
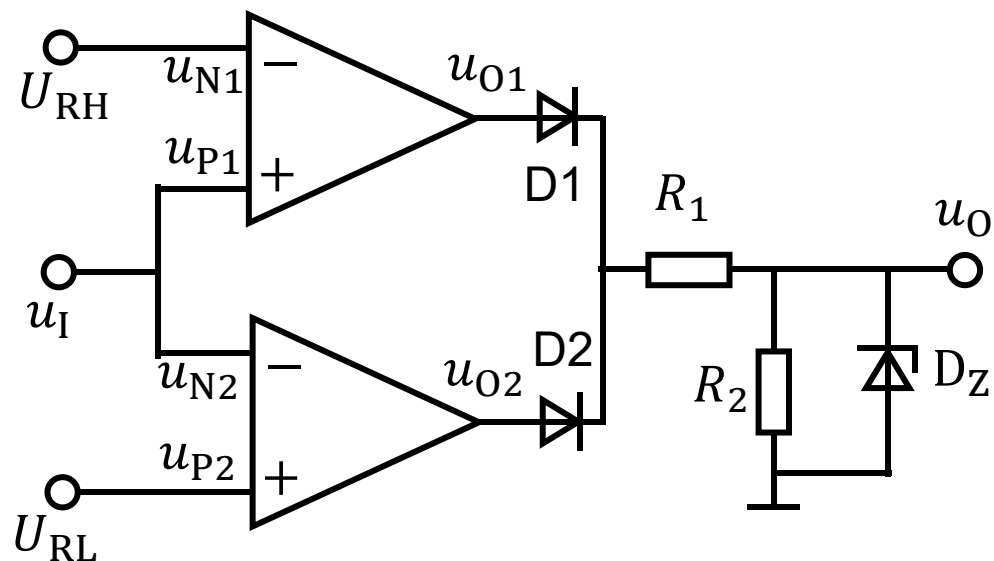
Working principle

(iii) When $U_{RL} < u_I < U_{RH}$

$$u_{O1} = -U_{OM} \quad \text{D1 is off}$$

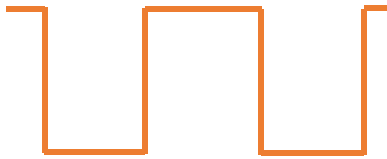
$$u_{O2} = -U_{OM} \quad \text{D2 is off}$$

$$u_O = 0$$



9.6 Non-sine-wave generator

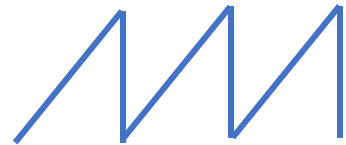
Rectangular wave



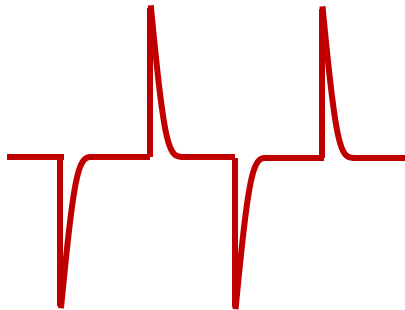
Triangular wave



Sawtooth wave

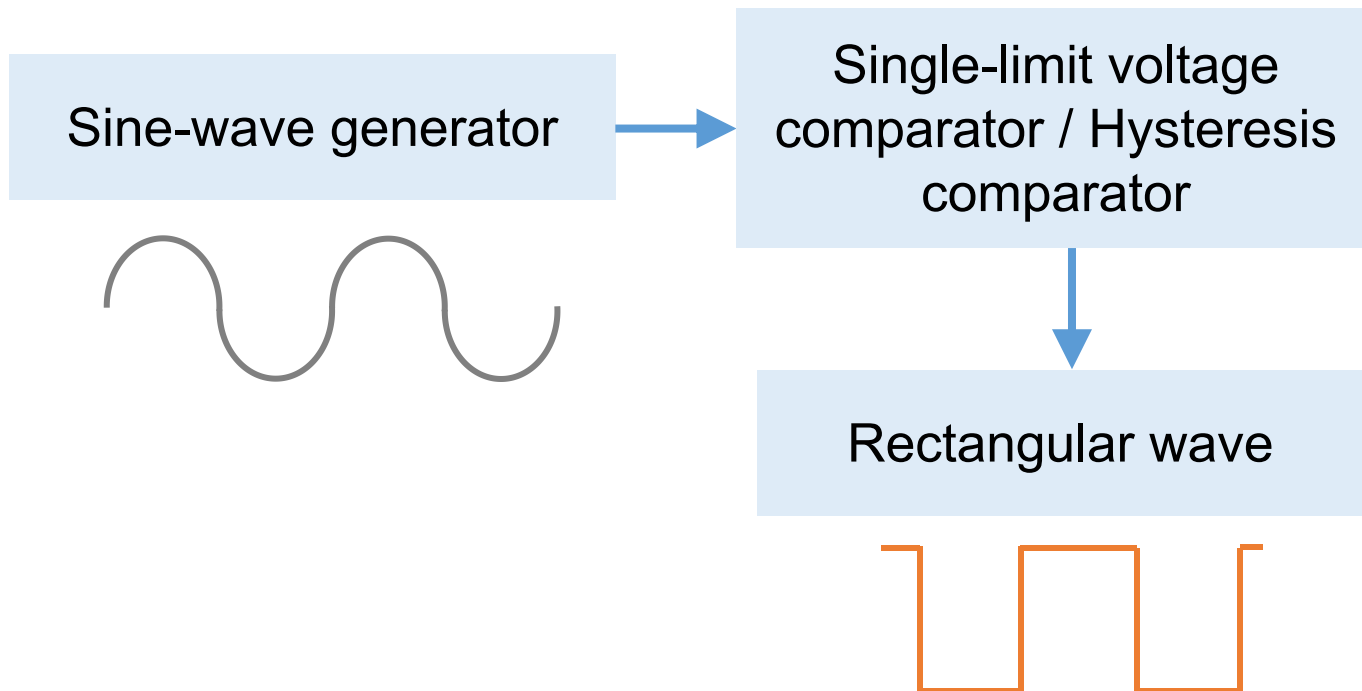


Peaked wave



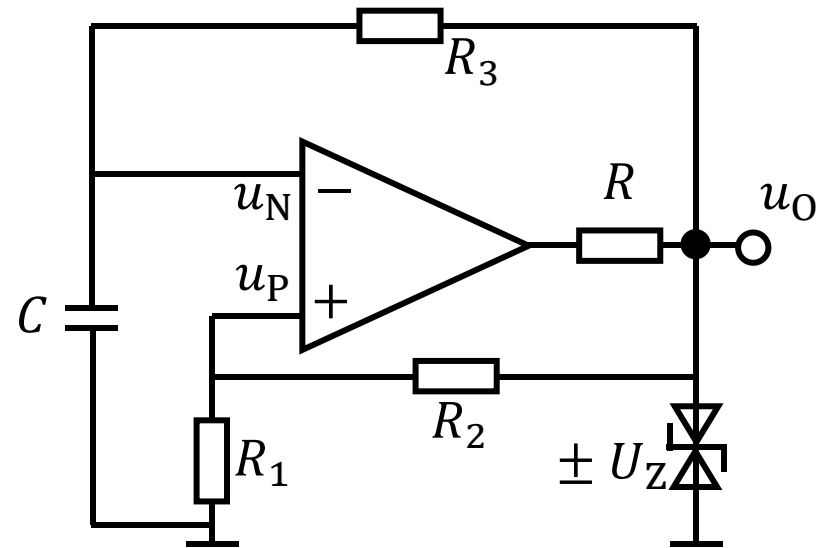
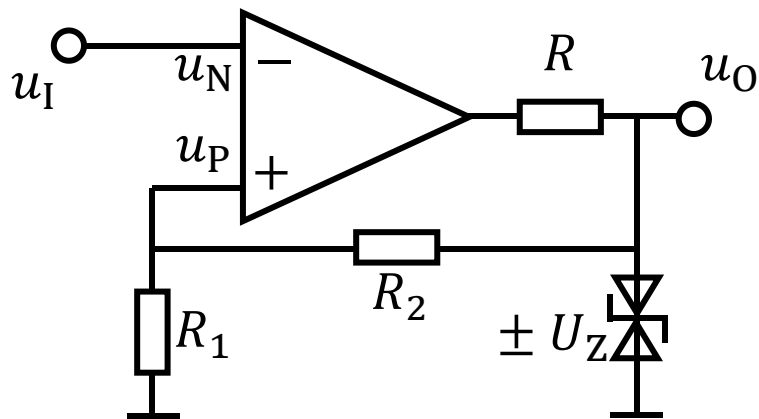
(1) Rectangular wave generator

(a) First method

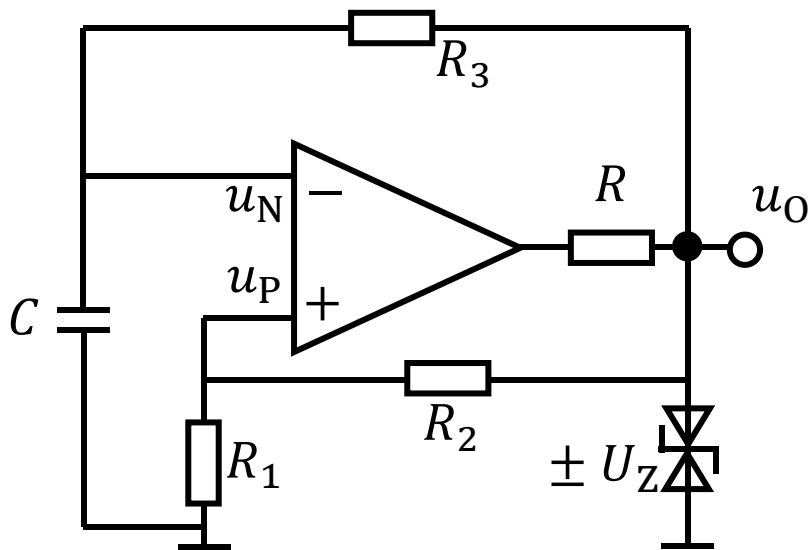


(b) Second method: use one operational amplifier

Hysteresis comparator



$u_C = u_N$ can be regarded as input voltage, what's the threshold voltage U_T ?



$u_C = u_N$ can be regarded as input voltage,
what's the threshold voltage U_T ?

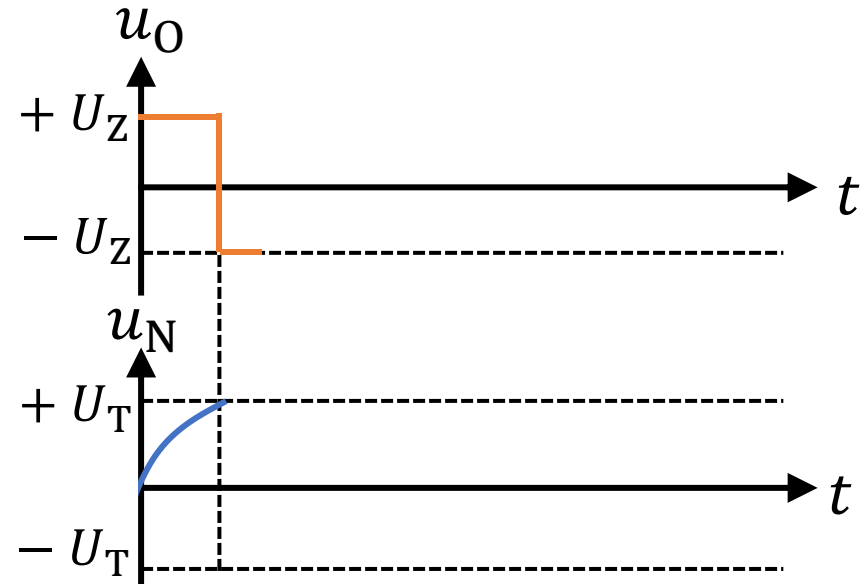
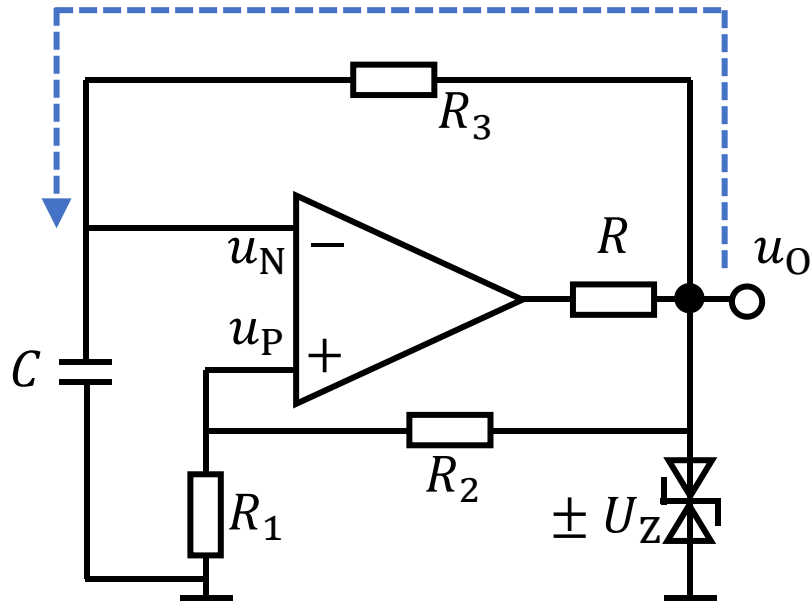
$$u_P - u_N = \frac{R_1}{R_1 + R_2} u_O - u_C$$

When $u_C = U_T$, $u_P - u_N = 0$

$$U_T = \frac{R_1}{R_1 + R_2} u_O$$

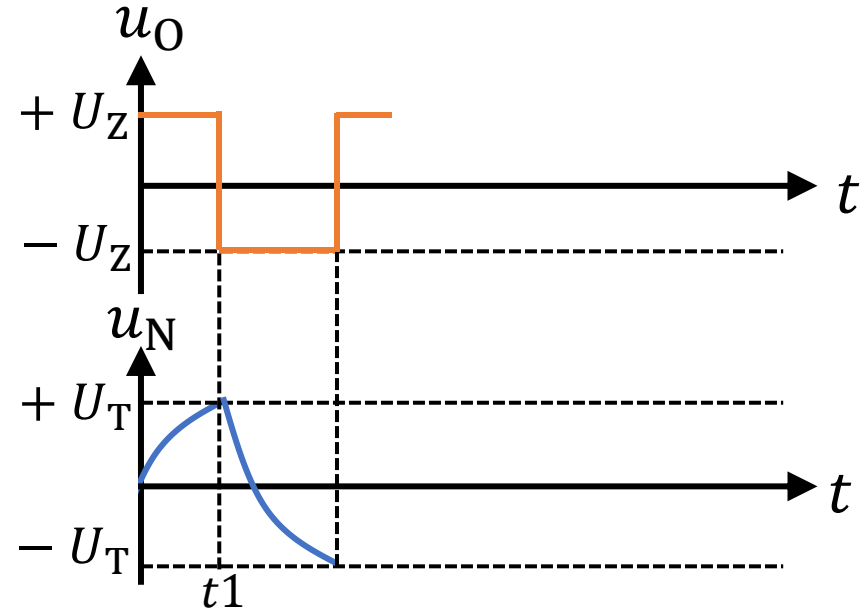
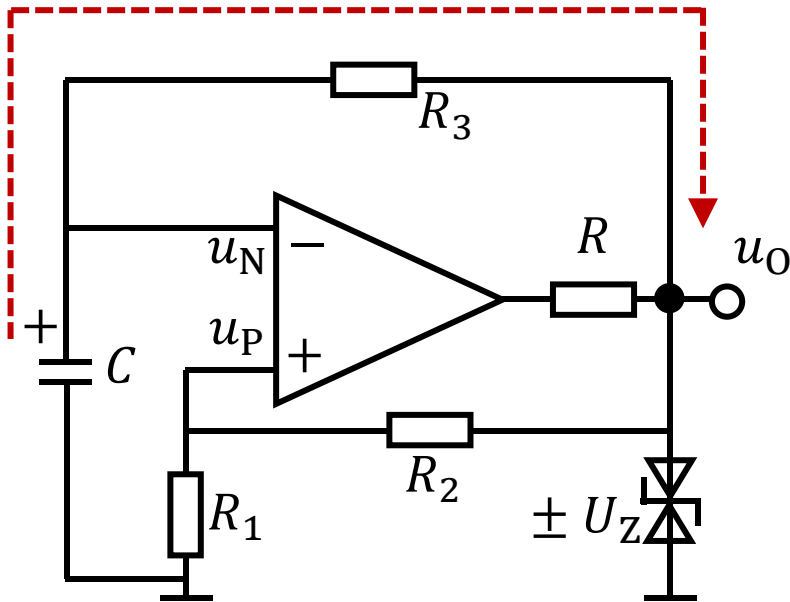
This means that when u_C (u_N) reaches U_T , the direction of u_O switches!!

Working principle



- (a) At $t = 0$, $u_O = +U_Z$:
- Fast** → Current flow through R_2 , R_1 : $u_P = +U_T$
 - Slow** → Current flow through u_O , R_3 , C , ground
- When $u_N = +U_T$, $u_O = -U_Z$ ← Charge C , u_N increase

Working principle



Fast

(b) At t_1 , $u_O = -U_Z$:

Current flow through R_2 , R_1 : $u_P = -U_T$

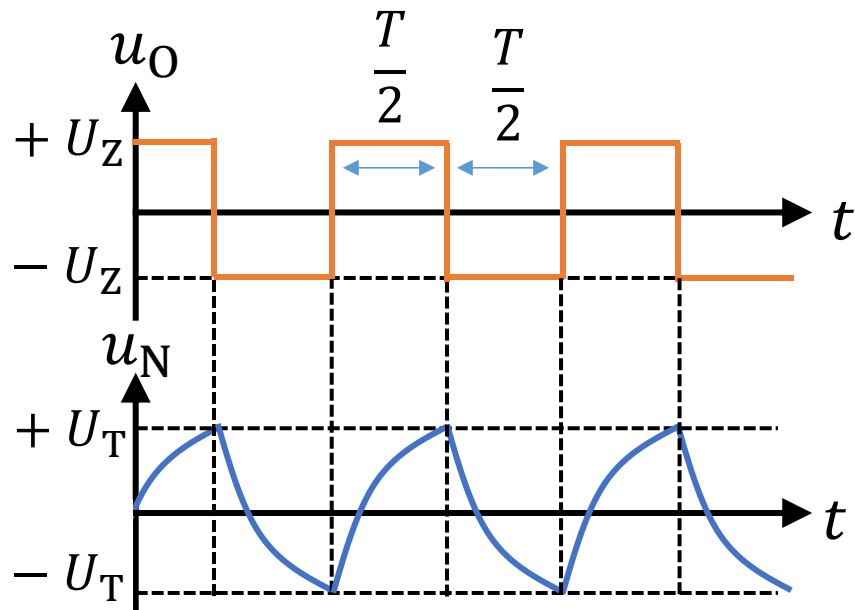
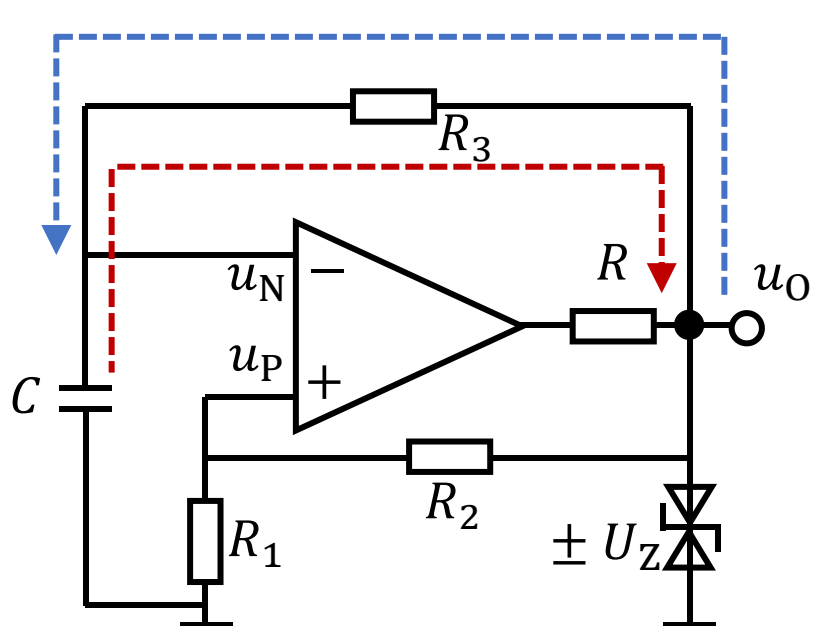
Slow

Current flow through ground, C , R_3 , u_O

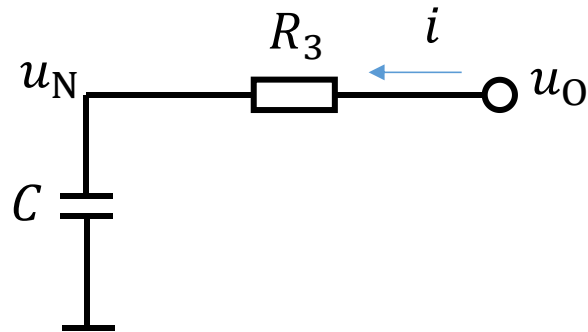
When $u_N = -U_T$, $u_O = +U_Z$

Discharge C , u_N decrease

Waveform of output signal u_O 输出波形



Oscillation period/frequency



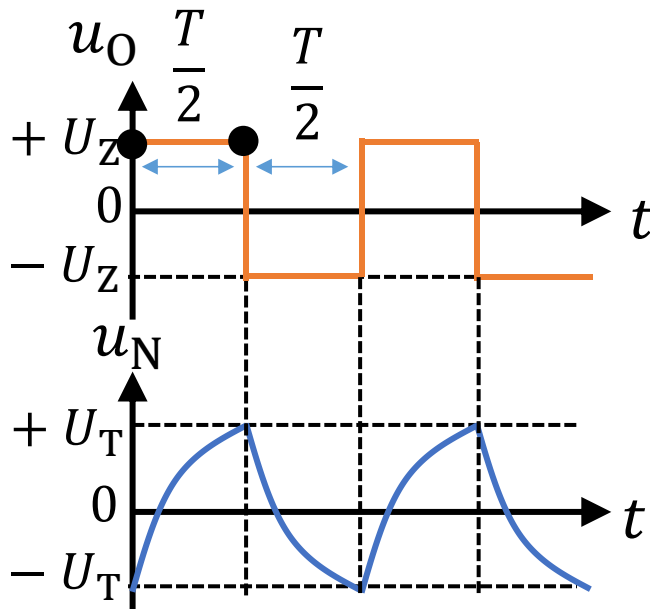
$$u_O = u_N + iR_3 \quad \Rightarrow \quad u_O = u_N + R_3 C \frac{du_N}{dt}$$

$$\text{Let } u_N = u_{N0} + Ae^{-t/B}$$

When $0 \leq t \leq T/2$:

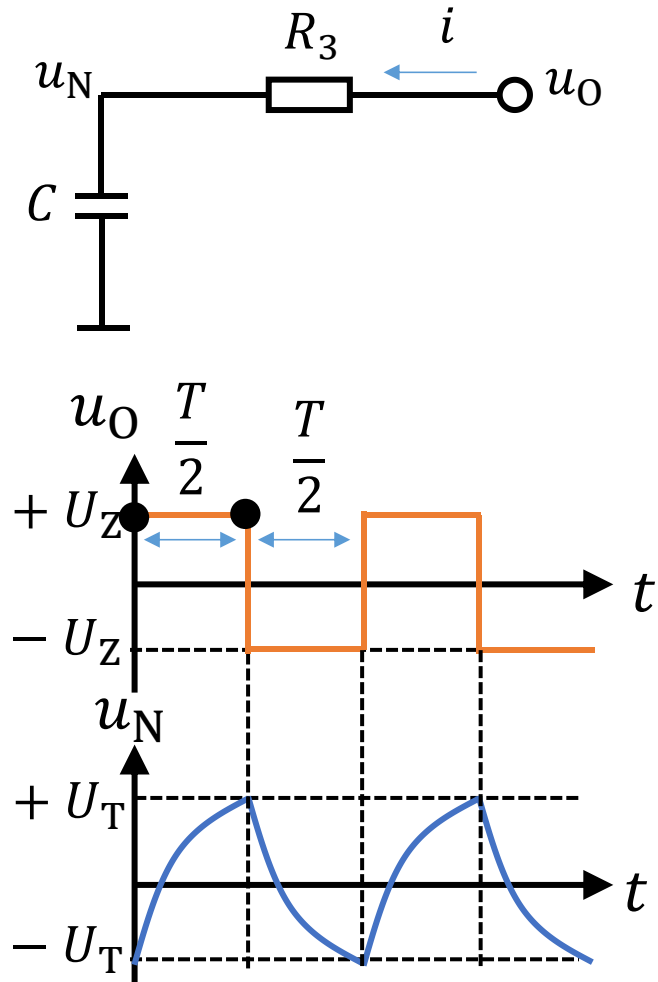
$$u_O = +U_Z$$

$$\left\{ \begin{array}{l} u_{N0} = U_Z \\ B = R_3 C \\ A = -(U_Z + U_T) \end{array} \right.$$



$$u_N(t) = U_Z - (U_Z + U_T)e^{-t/R_3 C}$$

Oscillation period/frequency



$$u_N(t) = U_Z - (U_Z + U_T)e^{-t/R_3C}$$

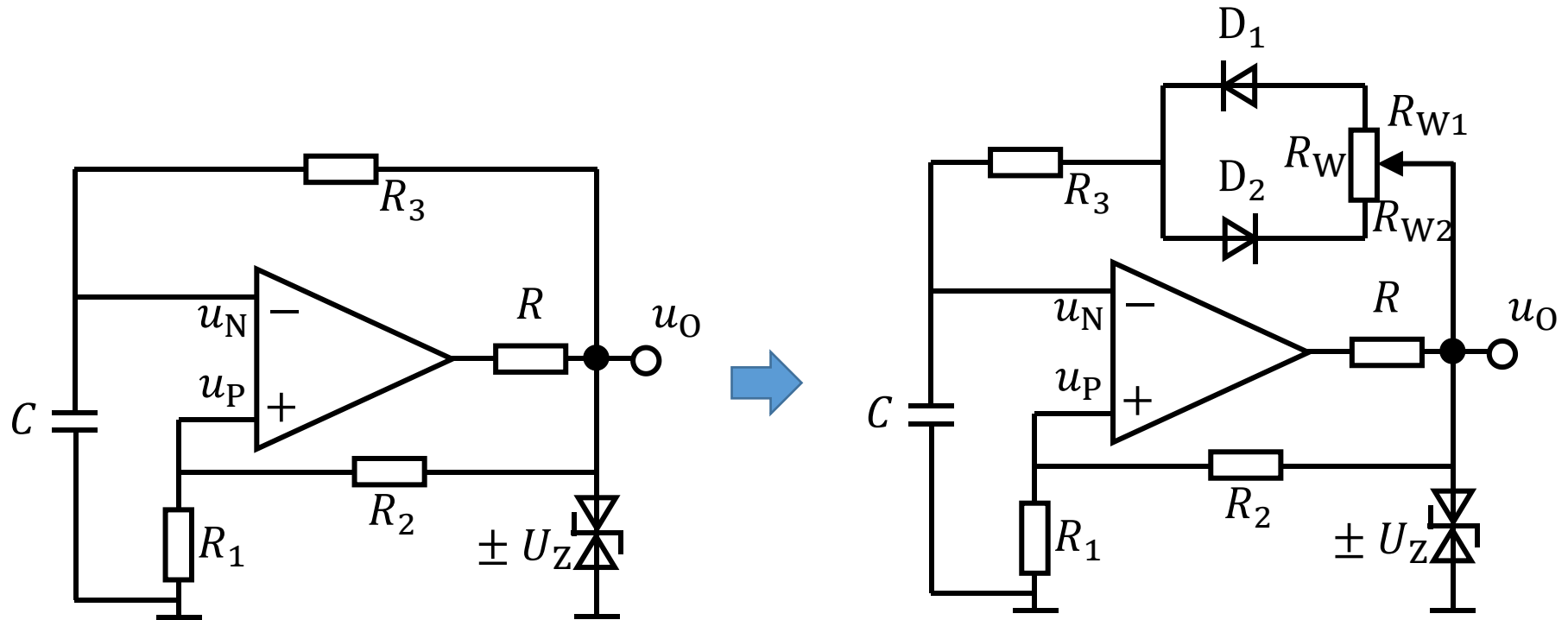
When $t = T/2$

$$U_T = U_Z - (U_Z + U_T)e^{-T/2R_3C}$$

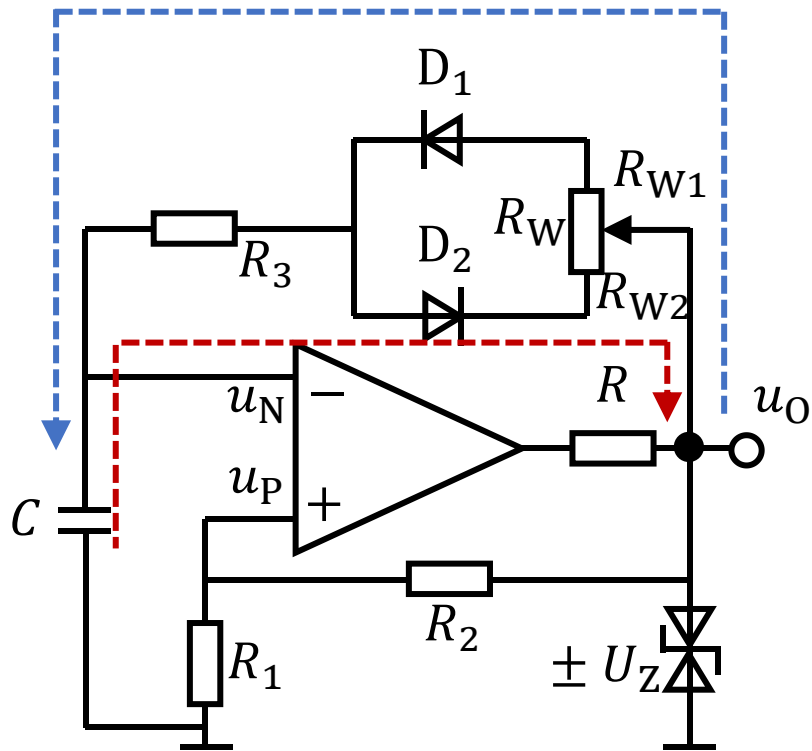
$$T = 2R_3C \ln\left(\frac{U_Z + U_T}{U_Z - U_T}\right)$$

$$= 2R_3C \ln\left(1 + \frac{2R_1}{R_2}\right)$$

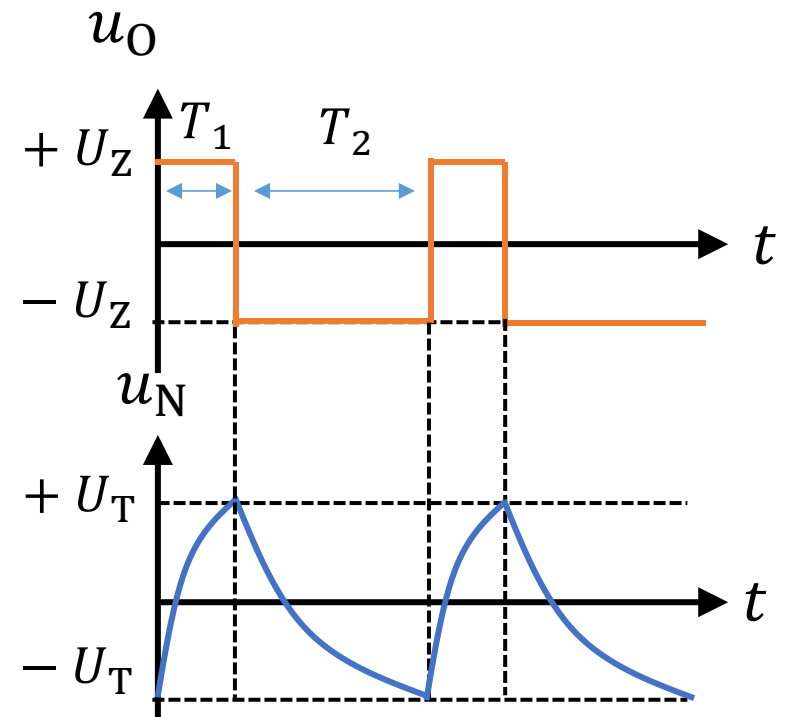
Question: what's the oscillation period of right circuit?



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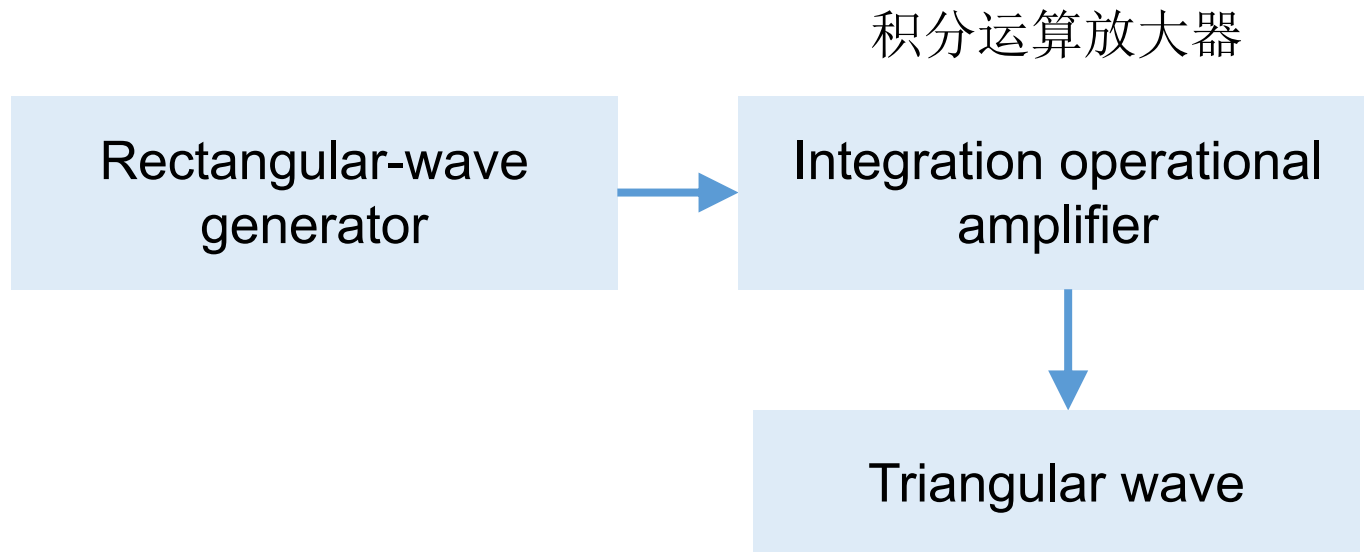
$$T_1 = (R_3 + R_{W1})C \ln\left(1 + \frac{2R_1}{R_2}\right)$$

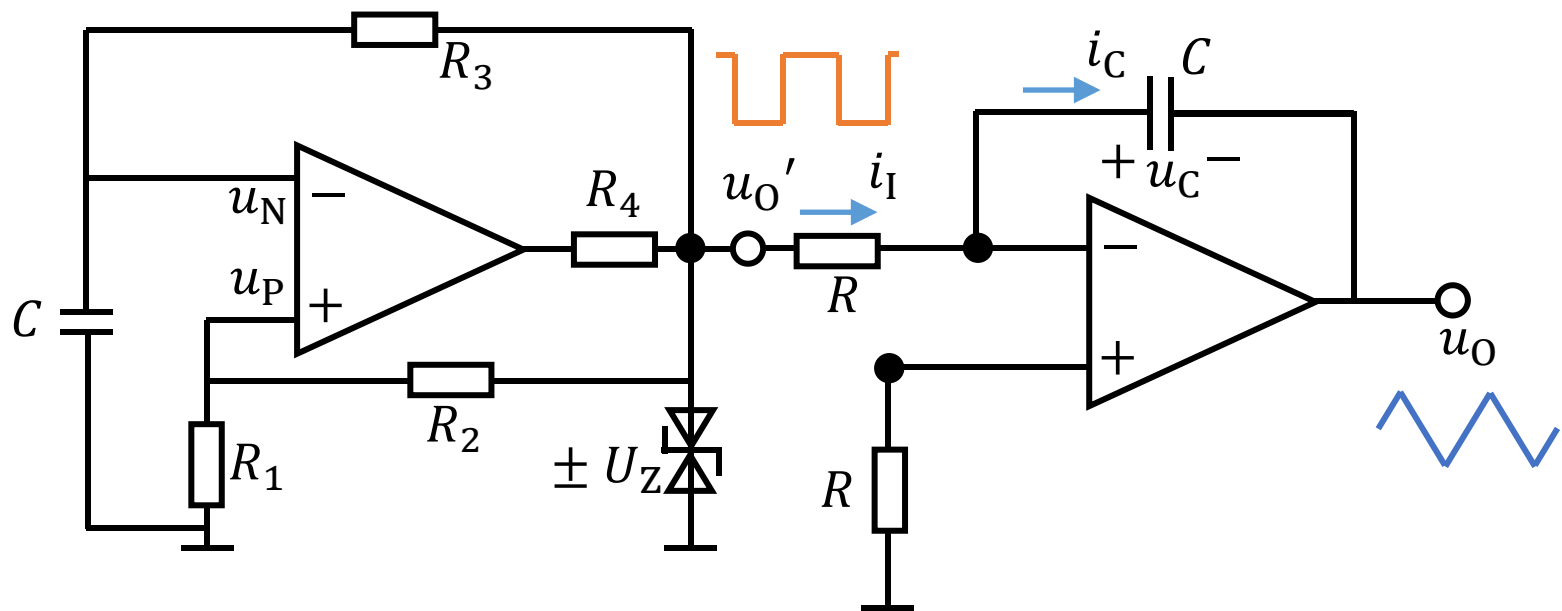


$$T_2 = (R_3 + R_{W2})C \ln\left(1 + \frac{2R_1}{R_2}\right)$$

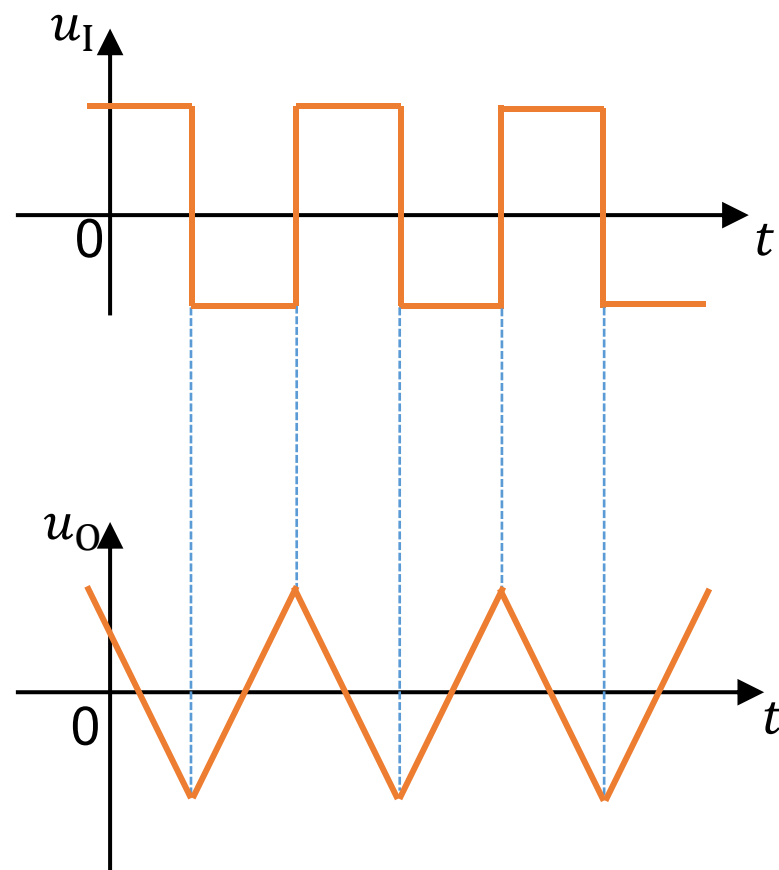
$$T = T_1 + T_2 = (2R_3 + R_W)C \ln\left(1 + \frac{2R_1}{R_2}\right)$$

(2) Triangular wave generator

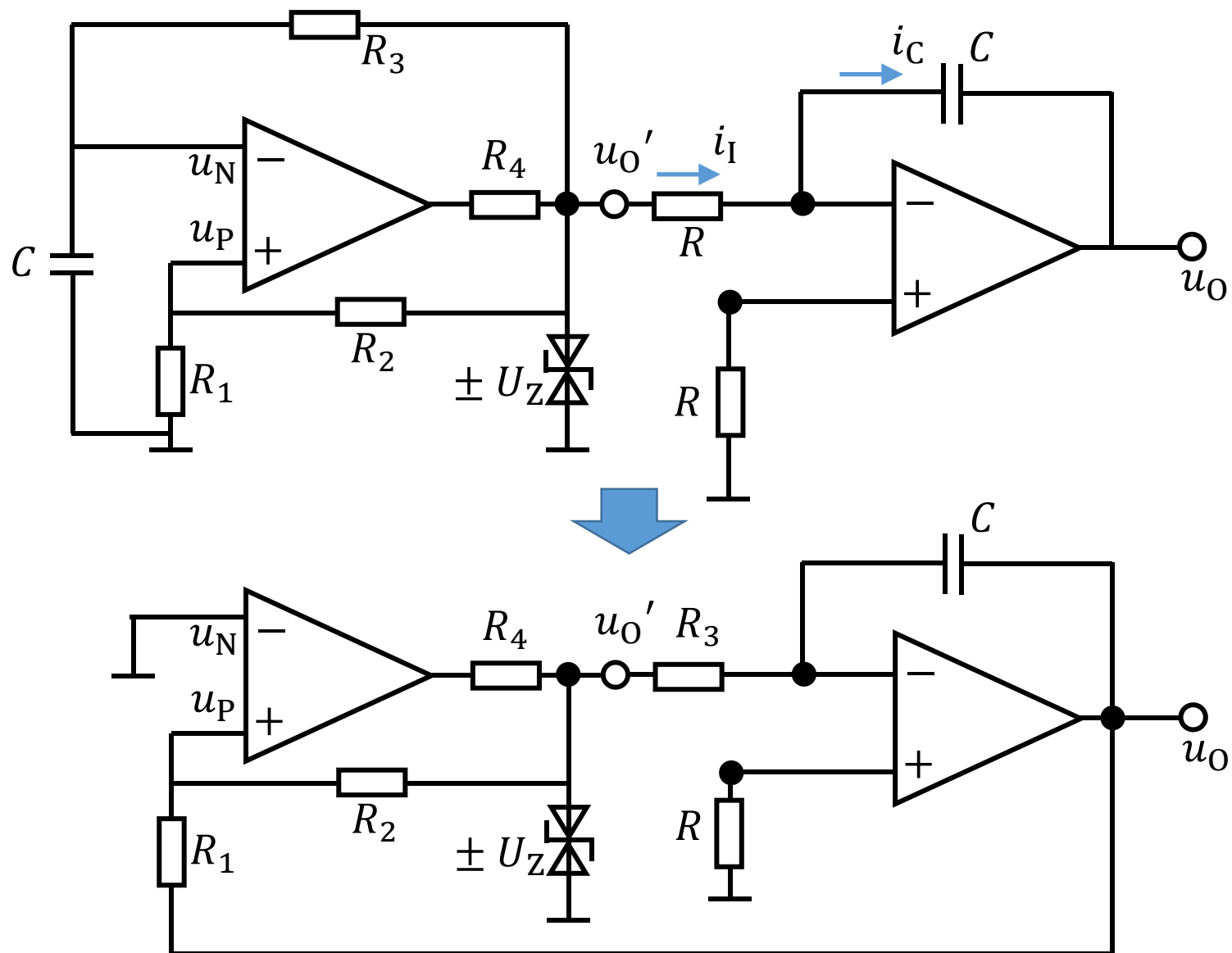


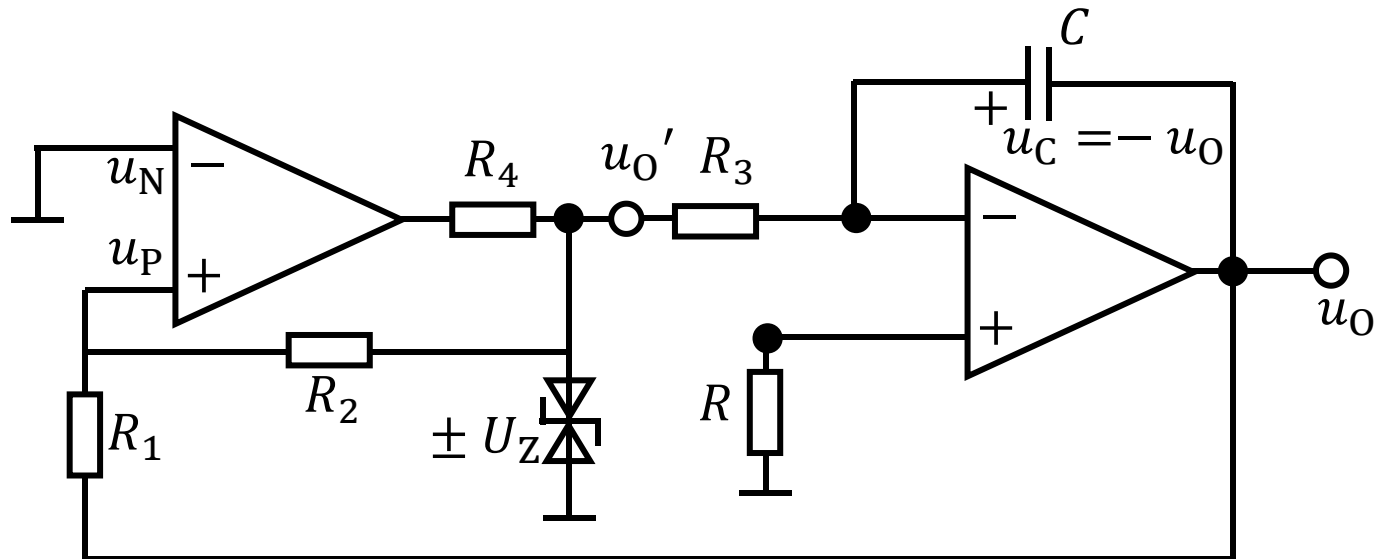


$$\left\{ \begin{array}{l} i_C = i_I = \frac{u_{O'}}{R} \\ i_C = \frac{dQ}{dt} = C \frac{du_C}{dt} \end{array} \right. \Rightarrow u_O = -u_C = -\frac{1}{C} \int i_C dt = -\frac{1}{RC} \int u_{O'} dt$$



Improved triangular wave generator

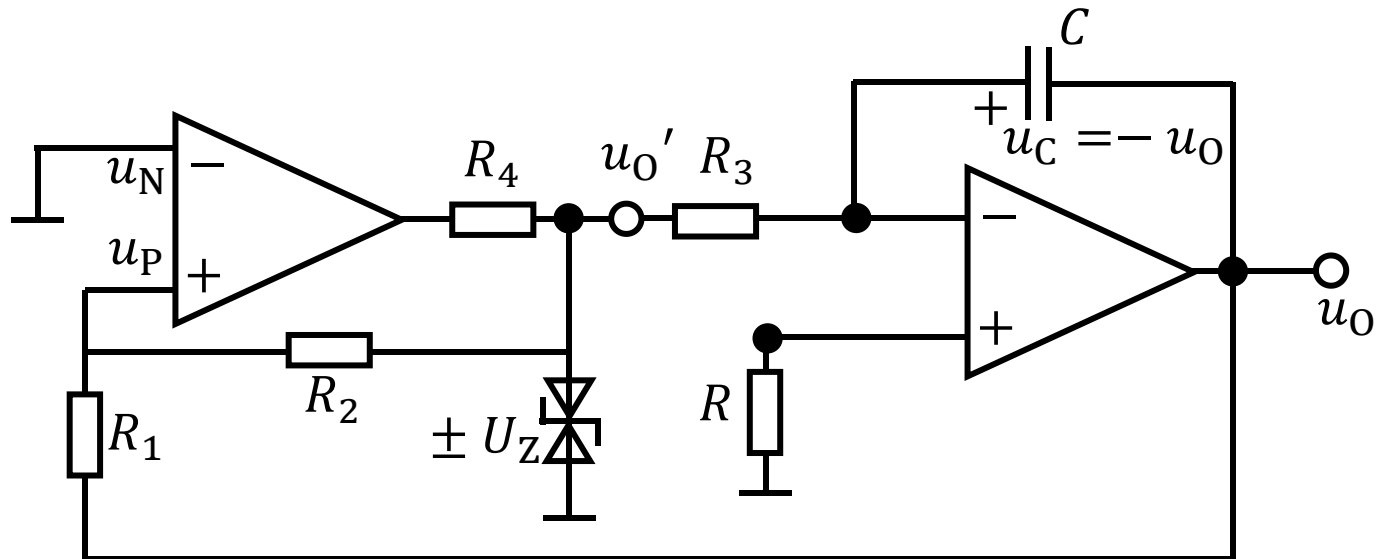




Voltage $u_O' = \pm U_Z$

$$u_P = \frac{R_2}{R_1 + R_2} u_O + \frac{R_1}{R_1 + R_2} u_O'$$

Question: u_C or u_O can be regarded as input voltage, what's the threshold voltage U_T ?

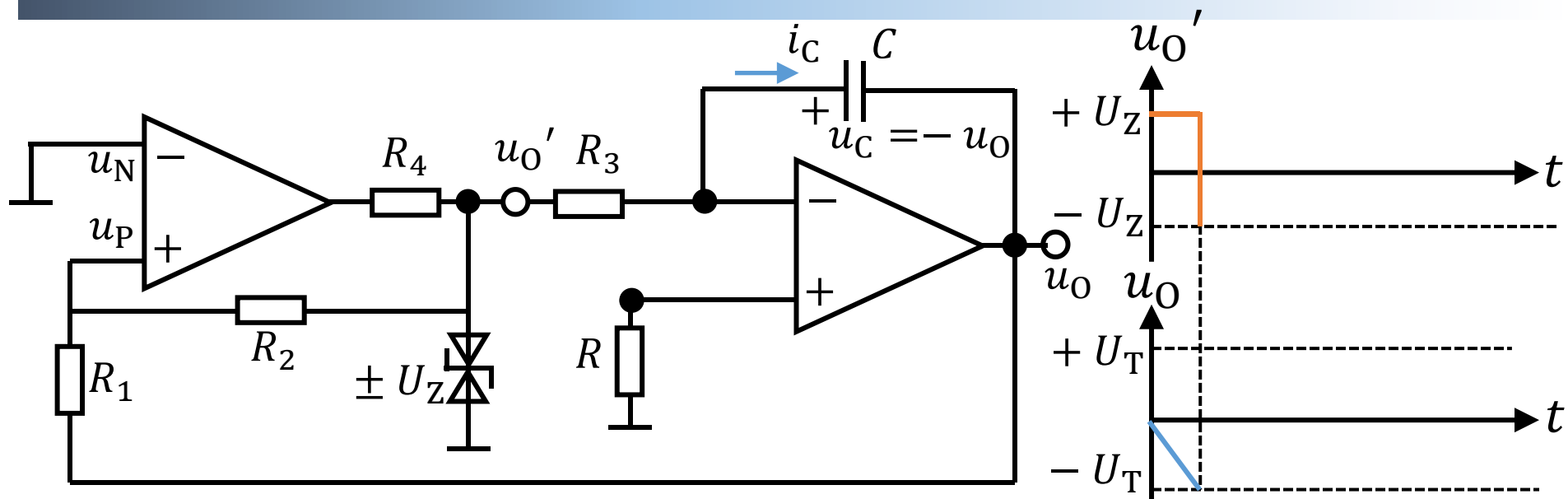


$$u_P = \frac{R_2}{R_1 + R_2} u_O + \frac{R_1}{R_1 + R_2} u_{O'}$$

At threshold voltage $u_C = -u_O = U_T$: $0 = -\frac{R_2}{R_1 + R_2} U_T + \frac{R_1}{R_1 + R_2} u_{O'}$

$$\Rightarrow U_T = \frac{R_1}{R_2} u_{O'}$$

$$\Rightarrow U_T = \pm \frac{R_1}{R_2} U_Z$$



$$u_P = \frac{R_2}{R_1 + R_2} u_O + \frac{R_1}{R_1 + R_2} u_{O'}$$

(a) At $t = 0$, $u_{O'} = +U_Z$,
 $u_O = 0$:

Fast

Current flow through $u_{O'}$, R_2 , R_1 : $u_P > 0$

Slow

Current flow through $u_{O'}$, R_3 , C

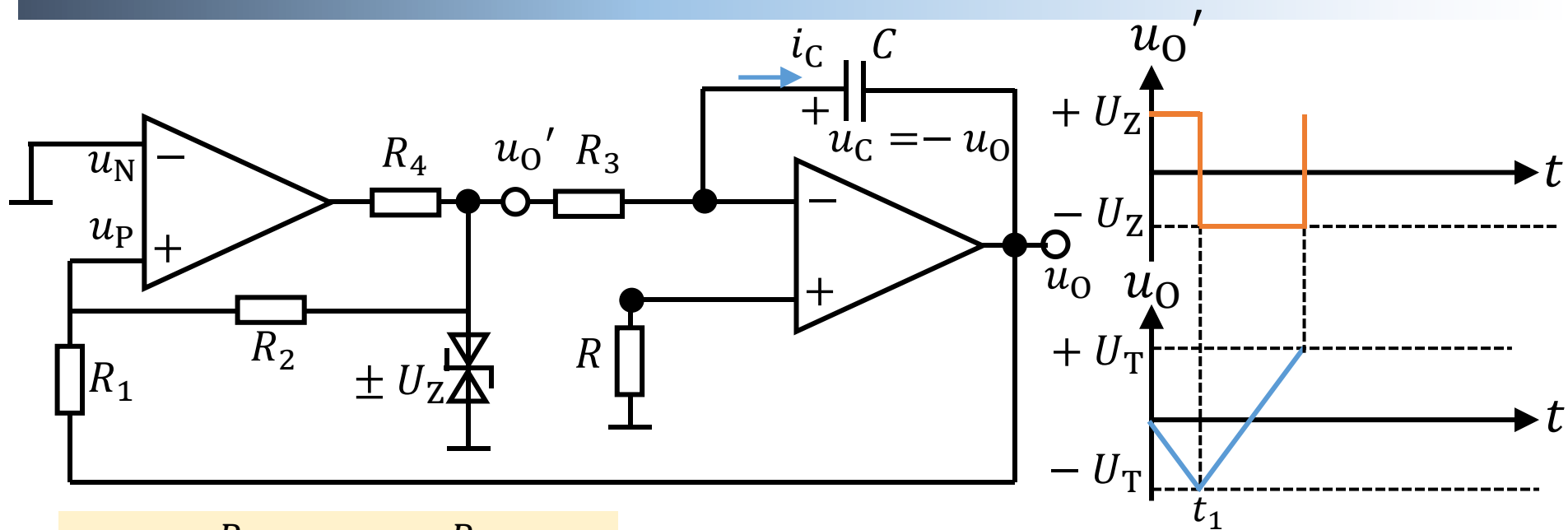


Charge C , u_C increase linearly with t

When $u_C = -u_O = +U_T$,
 $u_P = 0$, $u_{O'} = -U_Z$



$$u_C = \frac{1}{C} \int i_C dt = \frac{u_{O'}}{CR_3} \int dt = \frac{u_{O'}}{CR_3} t$$



$$u_P = \frac{R_2}{R_1 + R_2} u_O + \frac{R_1}{R_1 + R_2} u_{O'}$$

(b) At $t = t_1$, $u_{O'} = -U_Z$,
 $u_O = -U_T$:

Fast

Current flow through $u_{O'}$, R_2 , R_1 : $u_P < 0$

Slow

Current flow through $u_{O'}$, R_3 , C

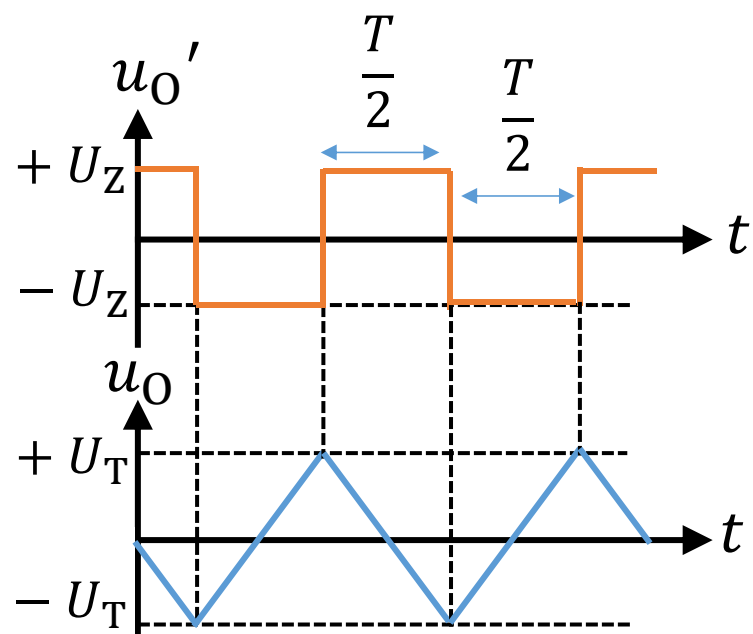
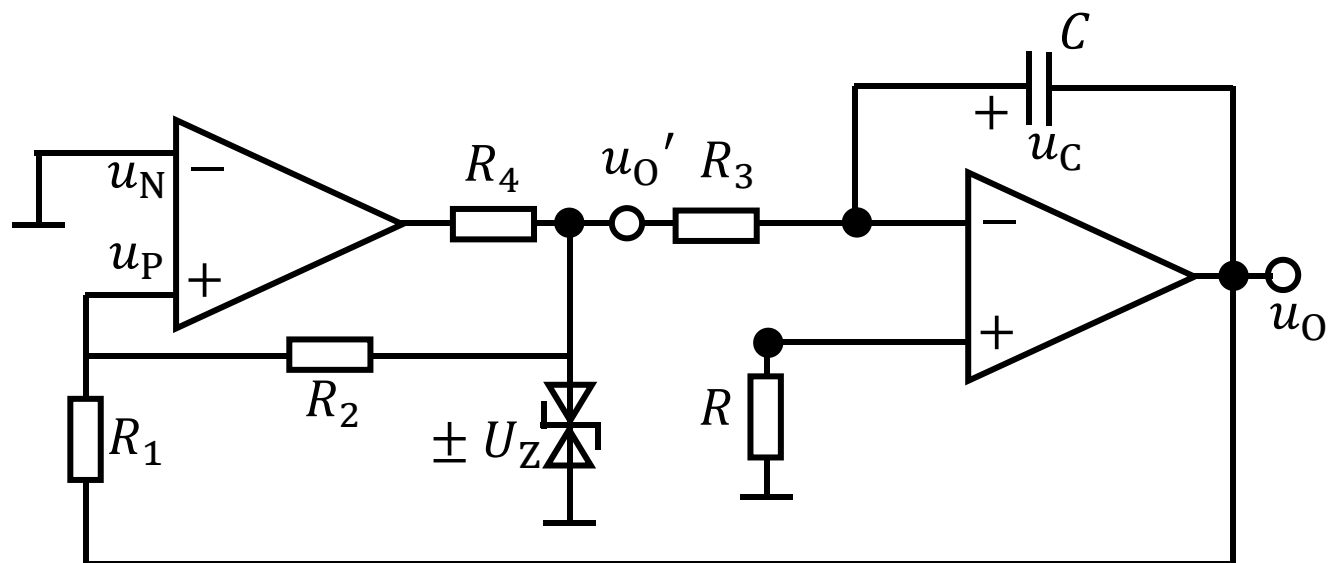


Discharge C , u_C decrease linearly with t

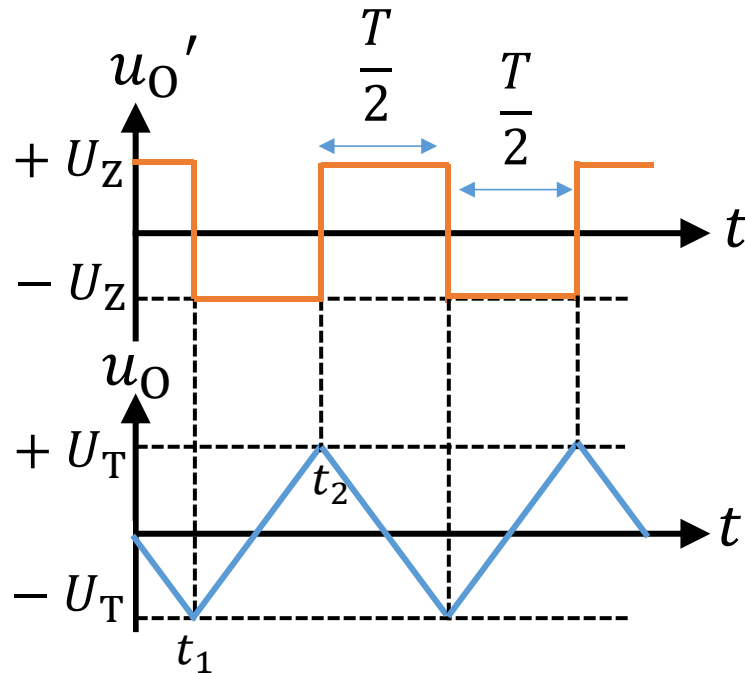
When $u_C = -u_O = -U_T$,
 $u_P = 0$, $u_{O'} = +U_Z$



$$u_C = U_T + \frac{1}{C} \int i_C dt = U_T + \frac{u_{O'}}{CR_3} (t - t_1)$$



Question: what's the oscillation period of right circuit?



When $t_1 < t < t_2$, $t_2 - t_1 = \frac{T}{2}$:

$$u_C = U_T + \frac{u_O'}{CR_3} (t - t_1)$$

$$-u_O = U_T + \frac{u_O'}{CR_3} (t - t_1)$$

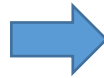
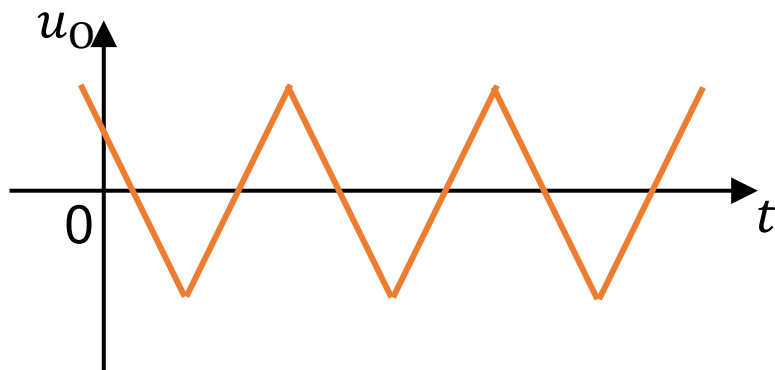
At $t = t_2$, $-U_T = U_T + \frac{-U_Z T}{CR_3} \frac{1}{2}$



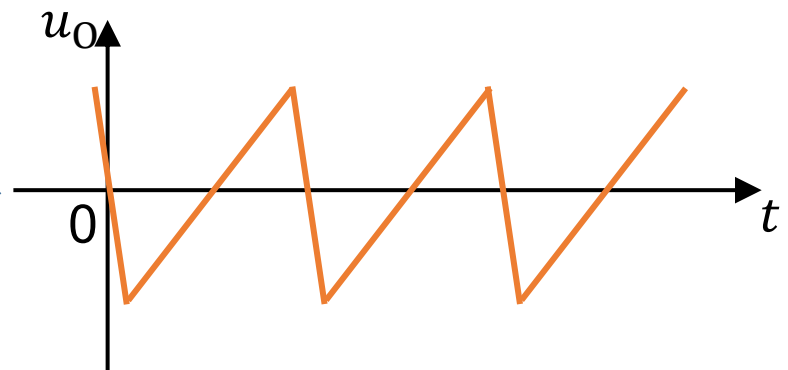
$$T = \frac{4U_T CR_3}{U_Z} = \frac{4R_1 R_3 C}{R_2}$$

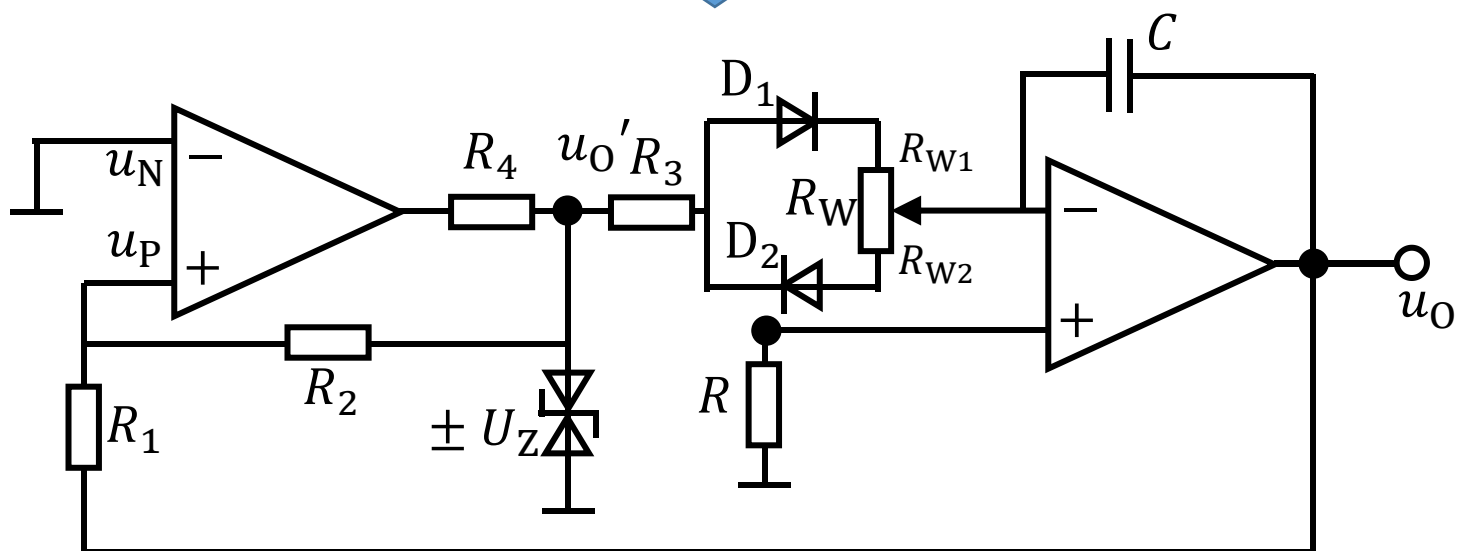
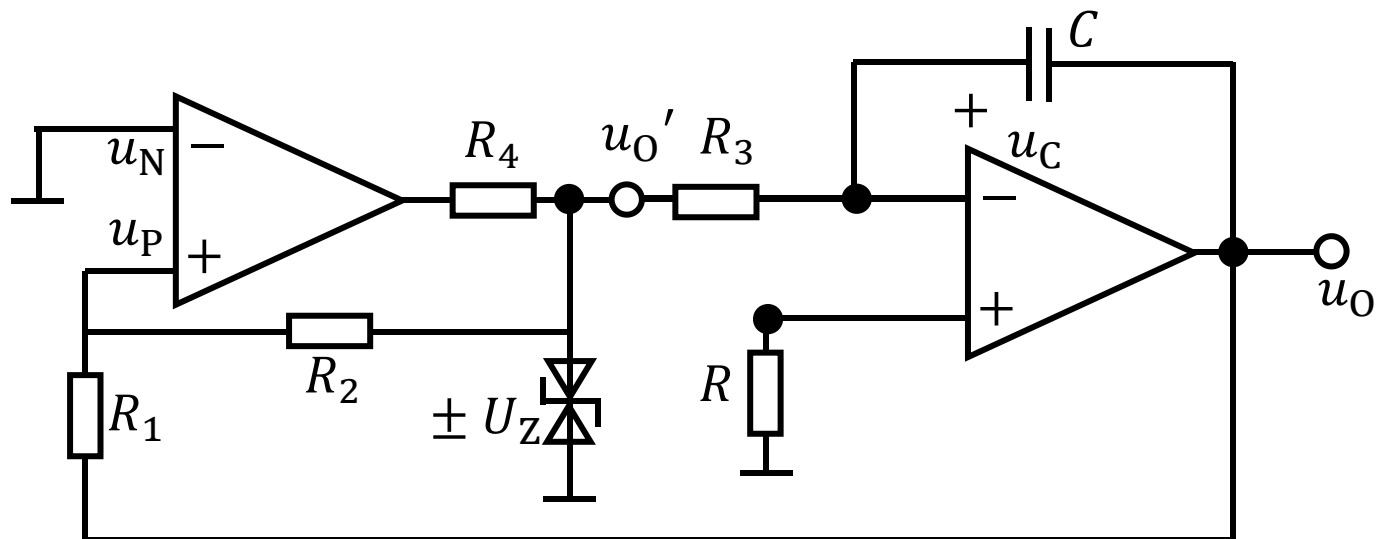
(3) Sawtooth wave generator

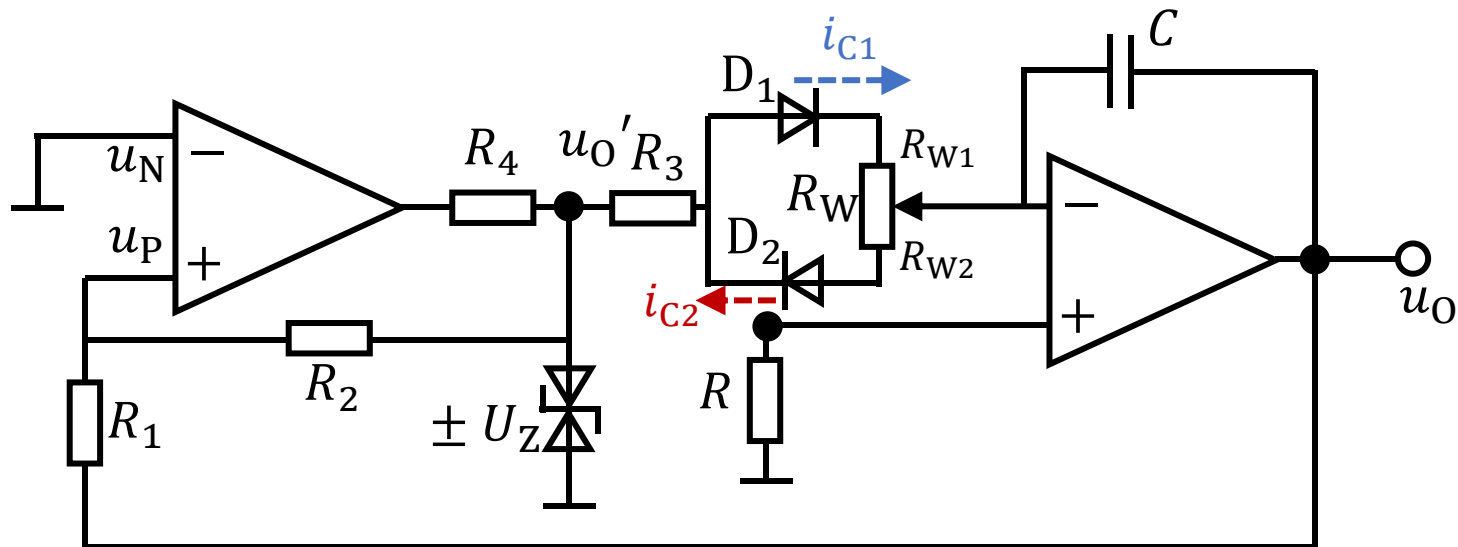
Triangular wave



Sawtooth wave







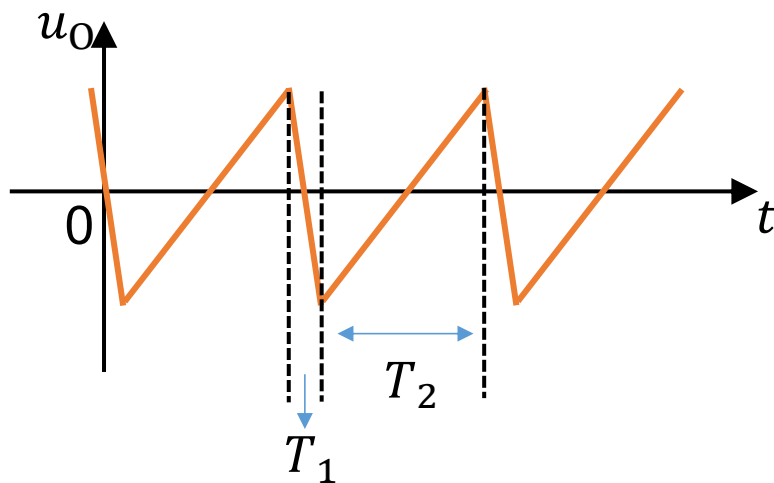
$$i_{C1} = \frac{U_Z}{(R_3 + R_{W1})}$$

$$i_{C2} = \frac{U_Z}{(R_3 + R_{W2})}$$

$$T_1 = \frac{2R_1(R_3 + R_{W1})C}{R_2}$$

$$T_2 = \frac{2R_1(R_3 + R_{W2})C}{R_2}$$

$$\text{Period: } T = T_1 + T_2 = \frac{2R_1(2R_3 + R_W)C}{R_2}$$



$$T_1 = \frac{2R_1(R_3 + R_{W1})C}{R_2}$$

$$T_2 = \frac{2R_1(R_3 + R_{W2})C}{R_2}$$

$$\frac{T_2}{T_1} = \frac{R_3 + R_{W2}}{R_3 + R_{W1}}$$

Homework 10: refer to QQ group and blackboard

第10次作业不强制要求提交。

